

XC24

The controller for constant current drivers

24 Channels

General Description

The XC24 is a local dimming controller featuring 24 channels to drive its companion chips, XD12 LED drivers that act as the voltage-based low side current sink for White LED back-lighting solution. This controller interfaces with XD12 drivers through one serial line to send and receive the data to be needed for local dimming.

The XC24 supports the SPI interface to communicate with the host, and the SPI interface can be daisy-chained, so it is effective to expand to support a large number of dimming areas.

For raising up dimming controllability, XC24 supports the FB signal to efficiently control the LED power supply voltage and two interrupt pins, nINT_FAULT to indicate the fault events like as the thermal status, the open detection, the short detection and the feedback status, and nINT_LD to indicate the room of internal buffers for dimming data.

The XC24 can support BIST functionality to ensure Back Light Unit is properly assembled.

Application

LED back-lighting with local dimming control

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Feature

- 24 channels local dimming controller for the XD12 LED drivers
- Flexible sizing for each channel
 - up to 31 daisy LED drivers per channel
 - up to 255 blocks per channel
 - up to 6,120 blocks per XC24
- SPI interface
 - up to 30MHz
 - support daisy-chain connection
- Configurable the baud-rate speed
- Frame synchronization support
- Fault events support of drivers through the interrupt function
 - Over-temperature event
 - Open detection
 - Short detection
 - FB status for VLED power efficiency
- TTL-level input range
- Supply voltage 4.5V ~ 5.5V
- 48-PIN QFN 6060 Package
- Latch-up Immunity ILATCH: JEDEC JESD78D Nov 2011: $\pm 100\text{mA}$
- ESD, HBM protection (Norm: MIL 883 E Method 3015 Human Body Model): $\pm 2\text{kV}$

Block Diagram

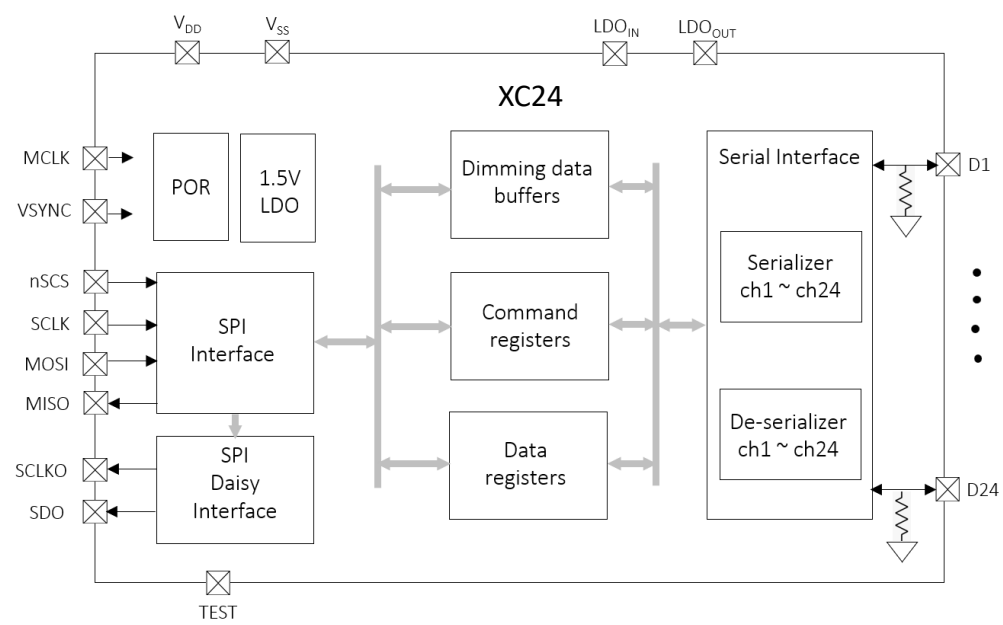


Fig. 1 Block Diagram

Application Circuit

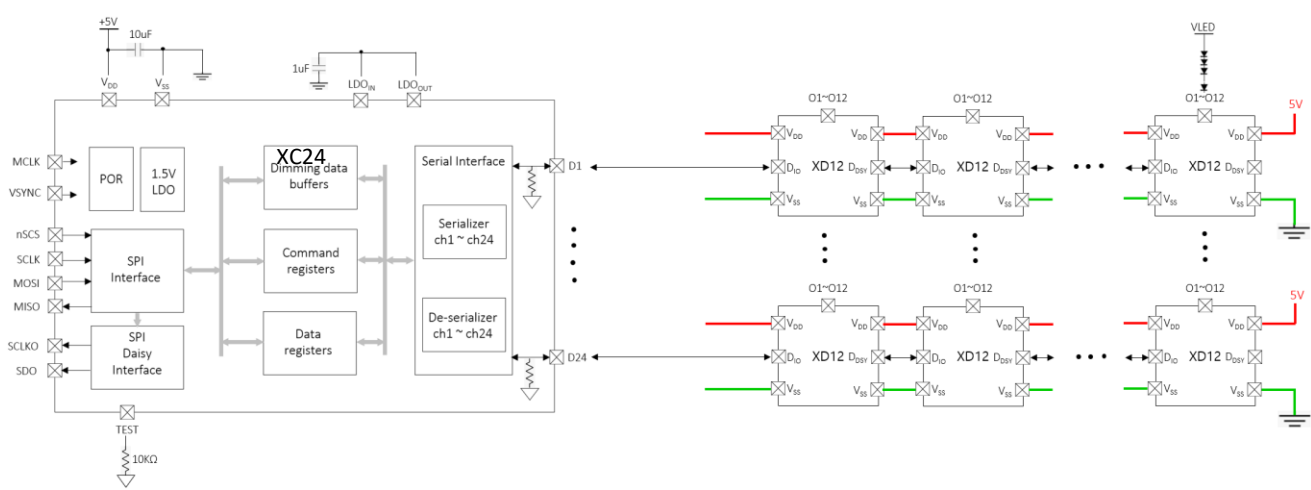


Fig. 2 Application Circuit

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Pin diagram (Top View)

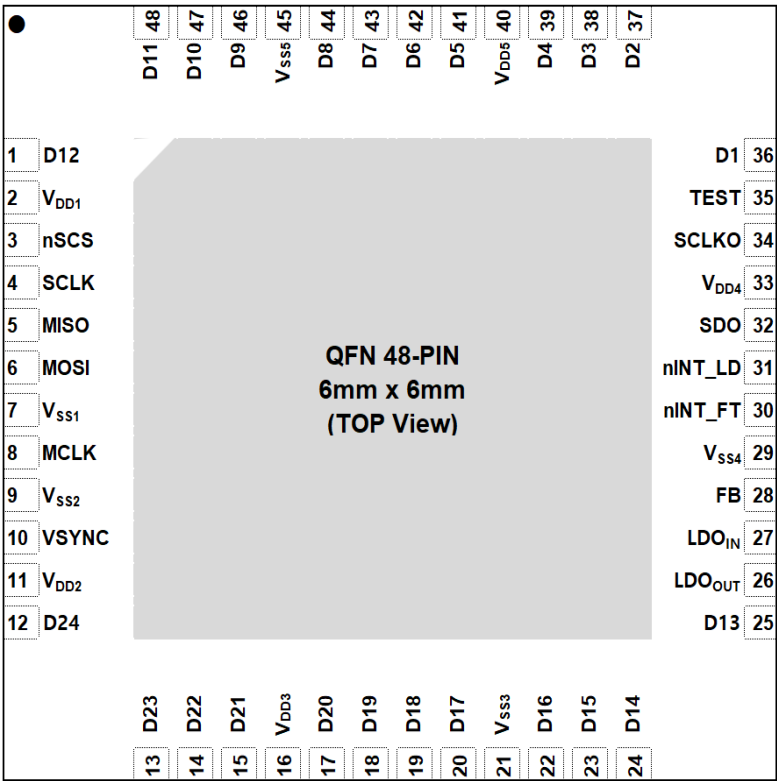


Fig. 3 Pin diagram

Pin description

Pin Name	Pin No.	I/O	Type	Description	Remark
D12	1	I/O	Digital	Data in/out	
V _{DD1}	2	I	Power	Digital V _{DD}	
nSCS	3	I	Digital	SPI Chip Select	
SCLK	4	I	Digital	SPI SCLK	
MISO	5	O	Digital	SPI MISO	
MOSI	6	I	Digital	SPI MOSI	
V _{SS1}	7	I	GND	Digital GND	
MCLK	8	I	Digital	Master Clock	

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Pin description

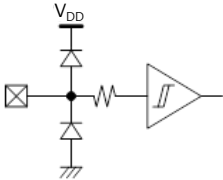
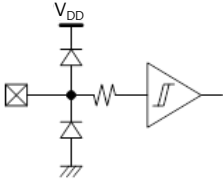
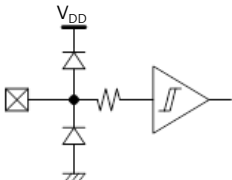
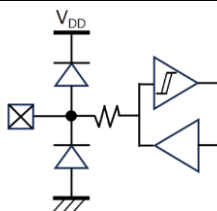
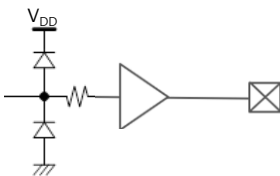
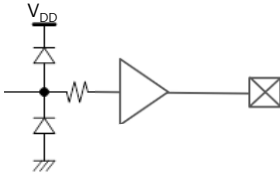
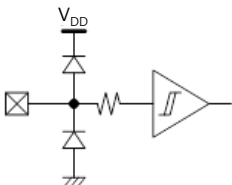
Pin Name	Pin No.	I/O	Type	Description	Remark
V _{SS2}	9	I	GND	Digital GND	
VS _{YNC}	10	I	Digital	Frame synchronization	
V _{DD2}	11	I	Power	Digital V _{DD}	
D24	12	I/O	Digital	Data in/out	
D23	13	I/O	Digital	Data in/out	
D22	14	I/O	Digital	Data in/out	
D21	15	I/O	Digital	Data in/out	
V _{DD3}	16	I	Power	Digital V _{DD}	
D20	17	I/O	Digital	Data in/out	
D19	18	I/O	Digital	Data in/out	
D18	19	I/O	Digital	Data in/out	
D17	20	I/O	Digital	Data in/out	
V _{SS3}	21	I	GND	Digital GND	
D16	22	I/O	Digital	Data in/out	
D15	23	I/O	Digital	Data in/out	
D14	24	I/O	Digital	Data in/out	
D13	25	I/O	Digital	Data in/out	
LDO _{IN}	26	I	Power	LDO feedback from LDO _{OUT}	Attach 1uF
LDO _{OUT}	27	O	Power	Internal LDO out to LDO _{IN}	Connect to LDO _{IN}
FB	28	O	Digital	Feedback for LED power supply	Open-drain, 10kΩ PU
V _{SS4}	29	I	GND	Digital GND	
nINT_FT	30	O	Digital	Interrupt for fault events	Open-drain, 10kΩ PU
nINT_LD	31	O	Digital	Interrupt for dimming data room	Open-drain, 10kΩ PU
SDO	32	O	Digital	SPI MOSI daisied out	Unused : 10kΩ PD
V _{DD4}	33	I	Power	Digital V _{DD}	
SCLKO	34	O	Digital	SPI SCLK daisied out	Unused : 10kΩ PD

Pin description

Pin Name	Pin No.	I/O	Type	Description	Remark
TEST	35	I	Digital	TEST mode pin	10Ω PD
D1	36	I/O	Digital	Data in/out	
D2	37	I/O	Digital	Data in/out	
D3	38	I/O	Digital	Data in/out	
D4	39	I/O	Digital	Data in/out	
V _{DD5}	40	I	Power	Digital V _{DD}	
D5	41	I/O	Digital	Data in/out	
D6	42	I/O	Digital	Data in/out	
D7	43	I/O	Digital	Data in/out	
D8	44	I/O	Digital	Data in/out	
V _{SS5}	45	I	GND	Digital GND	
D9	46	I/O	Digital	Data in/out	
D10	47	I/O	Digital	Data in/out	
D11	48	I/O	Digital	Data in/out	

Table 1. Pin Description

Equivalent Circuits of individual pins

Pad Name	I/O	Internal Circuit	Description
nSCS	I		SPI Chip Select
SCLK	I		SPI Clock Input
MOSI	I		SPI MOSI
MISO	O		SPI MISO
SCLKO	O		SCLK Daisy Out
SDO	O		MOSI Daisy Out
MCLK	I		Mater Clock

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Equivalent Circuits of individual pins

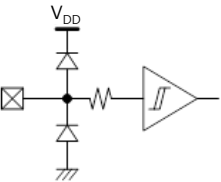
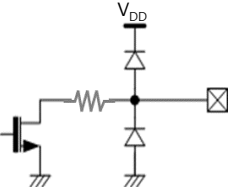
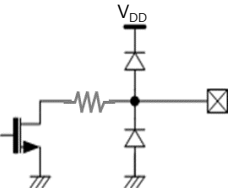
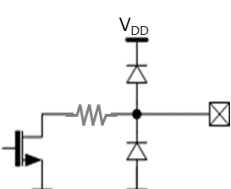
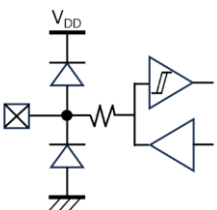
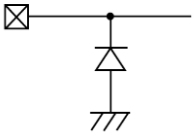
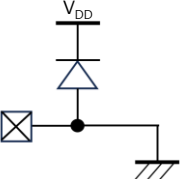
Pad Name	I/O	Internal Circuit	Description
VSYNC	I		Frame Sync.
nINT_FB	O		Interrupt for Fault (Open drain)
nINT_LD	O		Interrupt for LD (Open drain)
FB	O		Feedback for VLED (Open drain)
D	I/O		Data In/Out
VDD	-		Digital Power
VSS	-		Digital GND

Fig. 4 Equivalent Circuits of individual pins

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Maximum Absolute ratings (Ta = 25°C unless otherwise noted)

Parameter	Symbol	Value	Unit
Supply voltage	V _{DD}	-0.3 ~ 6.0	V
MCLK to GND voltage	V _{MCLK}	-0.3 ~ V _{DD} +0.5	V
VSYNCR to GND voltage	V _{VSYNCR}	-0.3 ~ V _{DD} +0.5	V
nSCSR to GND voltage	V _{SCSR}	-0.3 ~ V _{DD} +0.5	V
SCLK to GND voltage	V _{SCLK}	-0.3 ~ V _{DD} +0.5	V
MOSI to GND voltage	V _{MOSI}	-0.3 ~ V _{DD} +0.5	V
MISO to GND voltage	V _{MISO}	-0.3 ~ V _{DD} +0.5	V
SCLKO to GND voltage	V _{SCLK}	-0.3 ~ V _{DD} +0.5	V
SDO to GND voltage	V _{SDO}	-0.3 ~ V _{DD} +0.5	V
D voltage	V _{DIO}	-0.3 ~ V _{DD} +0.5	V
Ambient temperature	T _A	-20 ~ 85	°C
Junction temperature	T _J	-20 ~ 125	°C
Storage temperature	T _{STG}	-45 ~ 150	°C
Relative Humidity(non-condensing)	RH _{NC}	5 ~ 85	%
Moisture Sensitivity	MSL	Maximum Floor Life-time	3

Table 2. Maximum Absolute ratings

Notes : Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device

Electrostatic Discharge

Parameter	Symbol	Value	Unit	Remark
Electrostatic Discharge HBM ; Contact	ESDHBM	±2K	V	C=100 pF, R=1.5 kΩ
Electrostatic Discharge CDM ; Contact	ESDCDM	±500	V	R=1 Ω

Table 3. Electrostatic Discharge

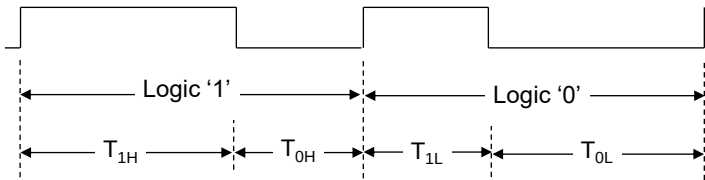
Electrical Characteristic (V_{DD}=5.0V, Ta=25.0°C unless otherwise noted)

Characteristics	Symbol	Condition	Value			Unit
			min	typ	max	
Supply						
Operating voltage	V _{DD}		4.5	-	5.5	V
Operating Current	I _{DD}		-	6.0	-	mA
Quiescent current	I _{DDD}		-	5.5	-	mA
MCLK						
Frequency	F _{MCLK}		1)	-	25	MHz
Duty			45	50	55	%
Jitter	△F _{MCLK}		-1		1	%
Input						
nSCS Input Voltage	V _{SCSL}	Chip Select, low level	-	-	0.8	V
	V _{SCSH}	Chip Select, high level	2.0	-	V _{DD}	V
SCLK Input Voltage	V _{SCLKL}	SPI Clock, low level	-	-	0.8	V
	V _{SCLKH}	SPI Clock, high level	2.0	-	V _{DD}	V
MOSI Input Voltage	V _{MOSIL}	MOSI, low level	-	-	0.8	V
	V _{MOSIH}	MOSI, high level	2.0	-	V _{DD}	V
VSYNC Input Voltage	V _{VSYNCL}	VSYNC low level	-	-	0.8	V
	V _{VSYNCH}	VSYNC high level	2.0	-	V _{DD}	V
Output						
MISO Output voltage	V _{MISOL}	MISO, low level	-	-	0.8	V
	V _{MISOH}	MISO, high level	2.0	-	V _{DD}	V
SCLKO Output voltage	V _{SCLKOL}	SCLKO, low level	-	-	0.8	V
	V _{SCLKOH}	SCLKO, high level	2.0	-	V _{DD}	V
SDO Output voltage	V _{SDOL}	SDO, low level	-	-	0.8	V
	V _{SDOH}	SDO, high level	2.0	-	V _{DD}	V
Bi-direction						
D _{IO}	V _{IL} V _{IH}	Input logic low level	-	-	0.5	V
		Input logic high level	3.5	-	V _{DD}	

Table 4. DC Electrical Characteristic

Electrical Characteristic (V_{DD} =5.0V, Ta=25.0°C unless otherwise noted)

Characteristics	Symbol	Condition	Value			Unit
			min	typ	max	
Bi-direction						
D _{IO}	V _{OL} V _{OH}	Output logic low level	-	-	0.5	V
		Output logic high level	V _{DD} - 0.5	-	V _{DD}	
	R _{PD}	Internal pull-down resistance	-	48	-	KΩ
Logic high time ²⁾	T _{1H} , T _{0L}		6	8	20	MCLK
Logic low time ²⁾	T _{1L} , T _{0H}		3	4	10	MCLK



Note : 1) It is recommended *the minimum MCLK should not be more than twice as slow as the SCLK* for stable fast-readout. If MCLK will be more than twice as slow as SCLK, then an appropriate latency is needed between the first 16 SCLKs for a command and the second 16 SCLKs for readout.

2) $T_{1H} \geq T_{0H} * 2$, $T_{0L} \geq T_{1L} * 2$, $(T_{1H} + T_{0H}) = (T_{0L} + T_{1L}) \leq 31$

Important statement

Global Technologies reserves the right to change the above-mentioned information without prior notice.

Power Sequence

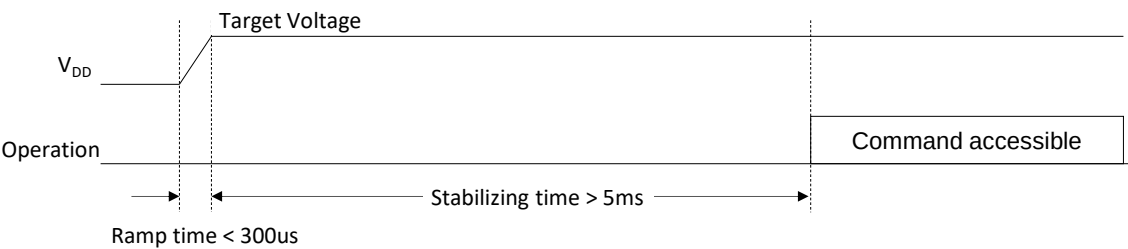


Fig. 5 Power up sequence timing

AC Characteristics (V_{DD S}=5.0V, Ta=25.0°C unless otherwise noted)

Characteristics	Symbol	Condition	Value			Unit	
			min	typ	max		
SCLK							
SCLK frequency	f _{CLK}		-	Typ ¹⁾	30	MHz	
SCLK duty	t _{CLK_DUTY}		48	-	52	%	
SCLK high time	t _{CH}		8	-	-	ns	
SCLK low time	t _{CL}		8	-	-	ns	
SCLKO							
SCLK to SCLKO delay	T _{CO_DELAY}		-	-	10.5 ⁽³⁾	ns	
The skew between SCLKO to SDO	t _{SKEW}		-	-	- 0.6 ⁽³⁾	ns	
SCLKO high time	t _{OCH}		t _{CH} - 1 ⁽³⁾	-	-	ns	
nSCS setup to the first rising SCLK	t _{CSS}		50	-	-	ns	
nSCS hold time from the last falling SCLK	t _{CSH}		3	-	-	MCLK	
nSCS high latency time	t _{CSHG}		3	-	-	MCLK	
MOSI setup time	t _{DS}		5	-	-	ns	
MOSI hold time	t _{DH}		2	-	-	ns	
MISO delay from the rising edge SCLK	t _{DO}		-	-	12	ns	
MISO high-Z from the nSCS de-assertion	t _{DOZ}		-	-	1	MCLK	

Note (3) It should be careful that the t_{CO_DELAY}, t_{SKEW}, t_{OCH} and t_{DO} are accumulated by the number of daisy devices if the daisy chain connection is used. So, for the read transaction, the proper timing should be considered in daisy chain.

Table 5. AC Electrical Characteristic

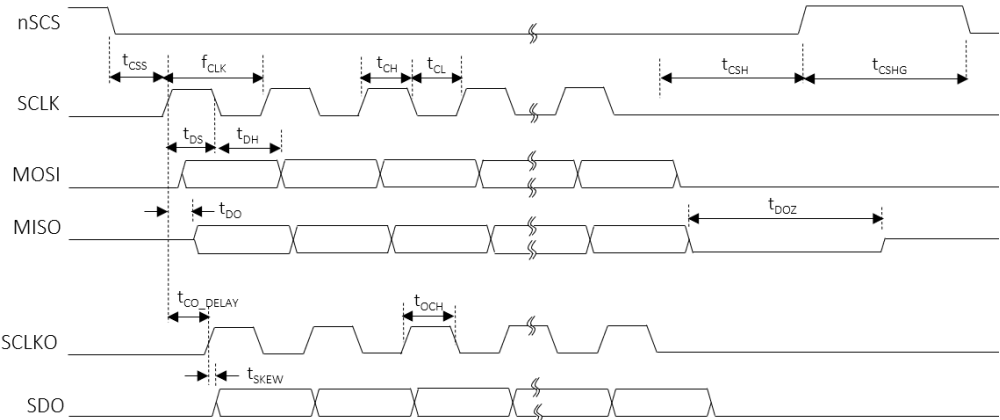


Fig. 6 SPI timing sequence

Functional Description

1. SPI Interface

For the local dimming data transmit a Serial Peripheral Interface is used. The SPI is configured to work as SPI slave. A Daisy Chain structure can be used to support large scale blocks array with several devices and operated up to 30MHz.

SPI Daisy Chain Structure

All SPI slaves share the Chip select signal, and these devices in Daisy chain can be treated as one big shift register. The devices can be enumerated as described below.

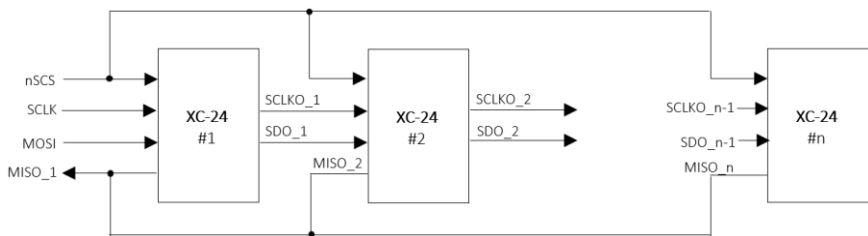


Fig. 7 SPI daisy connection

SPI Protocol

The XC24 has 3 types of the SPI Interface protocol, the register-read, the register-write and the LD data-transfer. The SPI protocol's format consists of a 16-bit command and a 16-bit register data or a series of 24(or 16)-bit LD data following the 16-bit command.

[Command format]

	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	Command code [1:0]		Address [7:0]								Burst size [5:0]					
Register-read	0	1	Start address [7:0]								Burst size [5:0]					
Register-write	1	0	Start address [7:0]								Burst size [5:0]					
LD-transfer	1	1	don't care								Line size [5:0]					

[Data format]

Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Data [15:0]															
Register-RW								Register data [15:0]							
LD1 transfer								LD_WIDTH [1:0] = 2'b00 : 24 bits /block							
LD2 transfer								LD_WIDTH [1:0] = 2'b01 : 24 bits /block							
LD3 transfer								LD_WIDTH [1:0] = 2'b10 : 24 bits /block							
LD4 transfer								LD_WIDTH [1:0] = 2'b11 : 16 bits /block							

Functional Description

SPI read/write sequence in burst

The SPI access sequence is as follows, depending on the type of access. There is no difference between a single and a burst access except for the number of data due to the burst size.

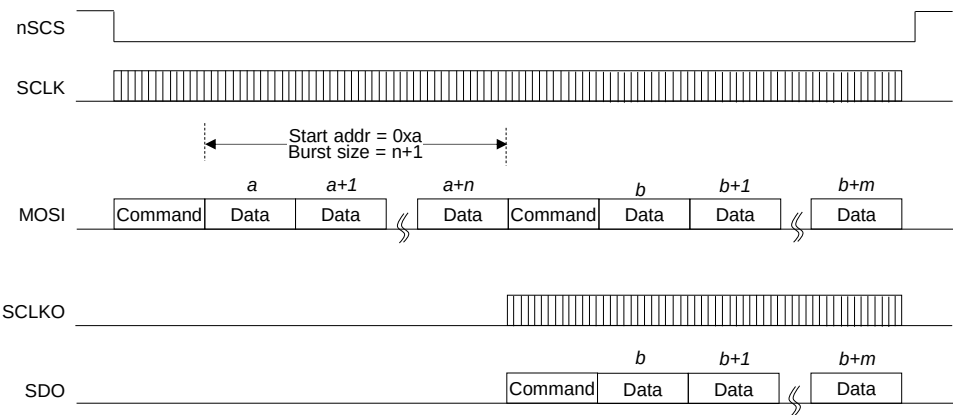


Fig. 8 The SPI Burst write in daisy-chained 2 devices

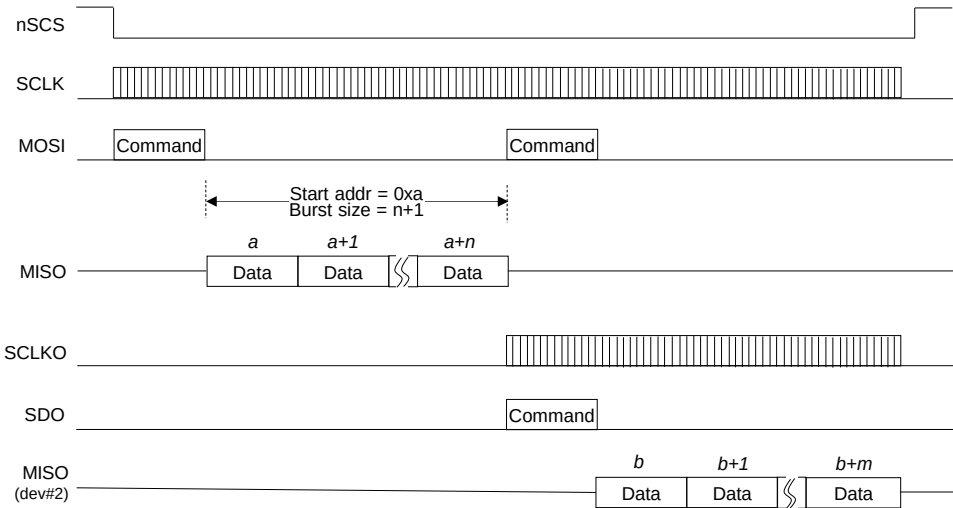


Fig. 9 The SPI Burst read in daisy-chained 2 devices

Functional Description

SPI LD transfer sequence

The SPI sequence for the LD transfer is as follows. The LD width depends on the LD_WIDTH value of the CHANNEL ENABLE 2 register. If the LD_WIDTH = 2'b00, then the LD data size of one block zone transferred through the SPI becomes 24 bits consisted of PAM 12 bits and PWM 12 bits. Even if the transmitted width is 24 bits, the valid bits are actually 20 bits, PAM 10 bits + PWM 10 bits. Therefore, the data size per block to be transmitted to the daisied devices is 20 bits.

LD_WIDTH [1:0]	LD width	LD format	Valid bits
2'b00	24 bits	PWM 8-bit + PAM 16-bit	PWM 8-bit + PAM 14-bit
2'b01	24 bits	PAM 12-bit + PWM 12-bit	PAM 10-bit + PWM 11-bit
2'b10	24 bits	PAM 12-bit + PWM 12-bit	PAM 10-bit + PWM 12-bit
2'b11	16 bits	LD 16-bit	LD 16-bit

Table 6. LD data size according to LD_WIDTH

The burst size unit means the amount of LD data corresponding to one line and the one line of LD data corresponds to the total LD data of all activated channels, and the number of active channel is determined by the CH_SIZE value of the ENABLE CHANNEL 2 register. Here, the CH_SIZE value should be equal to the number of activated channels by the CHx_EN of the ENABLE CHANNEL 1/2 registers.

Burst size unit = 1-line LD data = CH_SIZE * LD data size

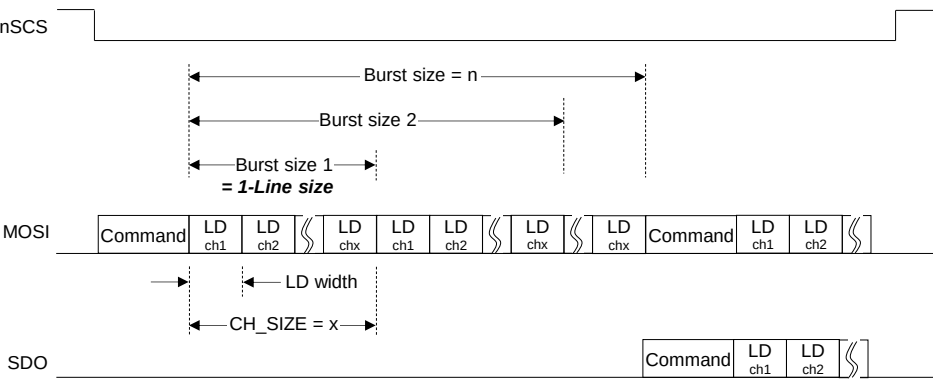


Fig. 10 The SPI LD transfer in daisy-chained 2 devices

Functional Description

2. Serial Interface

The XC24 has a total of 24 channels for serial daisy connections with a total of 31 XD12 devices. Each channel can transmit the data to the XD12 and receive the data from the XD12 through one line daisy to the XD12 devices. The XD24 has a total of 5 instructions for transferring the data to the XD12 and 3 instructions for receiving the data from the XD12.

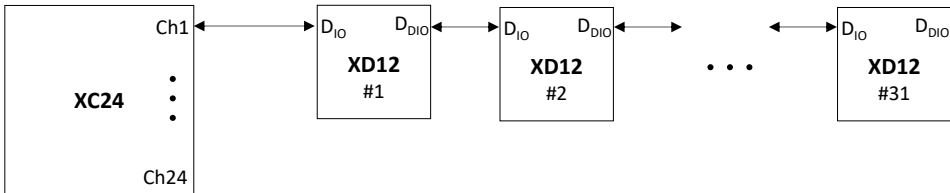


Fig. 11 The daisy scheme

The commands are listed as shown in the Table 2. and described in more detail below.

Instruction	Transaction direction		Description
	Command	Data	
Global write	XC → XD	XC → XD	Global-write command the same register data between the XD devices
Local write	XC → XD	XC → XD	Local-write command the different register data between the XD devices
LD transfer	XC → XD	XC → XD	Local Dimming data-transfer command to the XD devices
ID gen	XC → XD	-	ID generation command of the XD devices in daisy each channel
Sync gen	XC → XD	-	Sync-generation command to the XD devices
Sync auto-gen	XC → XD	-	Automatically sync-generation command sync on the XD devices
Local read	XC → XD	XD → XC	Local-read command to the XD devices and read-back from the XD devices
Fault read	XC → XD	XD → XC	Fault-read command to the XD devices and read-back from the XD devices
Fault auto-read	XC → XD	XD → XC	Automatically fault-read command to the XD devices and read-back from the XD devices

Table 7. Instruction list

Functional Description

ID-generation

The ID-GEN COMMAND register is used to set a unique ID for each device and each device's ID helps identifying when each device should react on a read-back operation. Therefore, after POR this command should be transacted at the first time for proper operation of the XD12 devices before transacting other commands. If the START bit of ID-GEN COMMAND register is set, then all activated channels begin issuing the command signals to all XD12 daisied in each channel and the command signal streams repeat continuously for the number of daisied devices.

The number of daisied devices is determined by the DAISY SIZE register of each channel. After the command signal streams are completed, all devices will have their own ID respectively. The ID is numbered in opposite direction of daisied line. It is careful that the writing data is transferred from the first located device, the last number ID, to the last located device, the first number ID, in daisied line but the reading data is received from the last located device to the first located device sequentially.

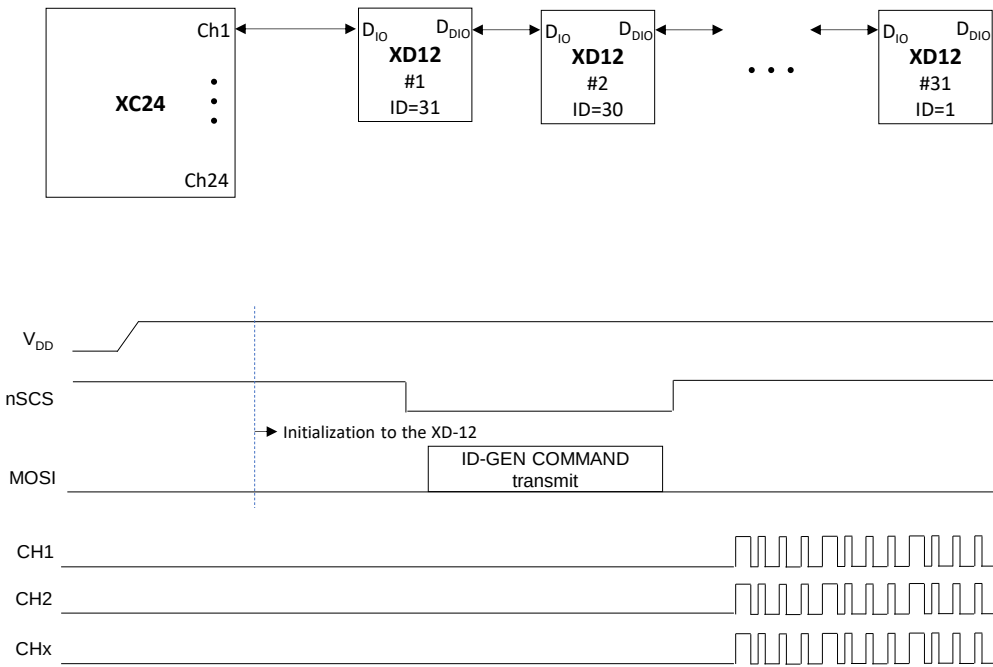


Fig. 12 The sequence of ID-GEN command

Functional Description

Global-write

The Global-write command is used to write the register data to the XD12 only if all XD12 devices daisied to a channel of the XC24 have the same register data. If the register data to be written to the XD12 are different each other, at that time the Local-write command should be used. For executing the Global-write command properly, the GLOBAL-WRITE DATA register should be firstly written with desired value and then the START bit of the GLOBAL-WRITE COMMAND register should be set with desired address in order to begin transacting this command.

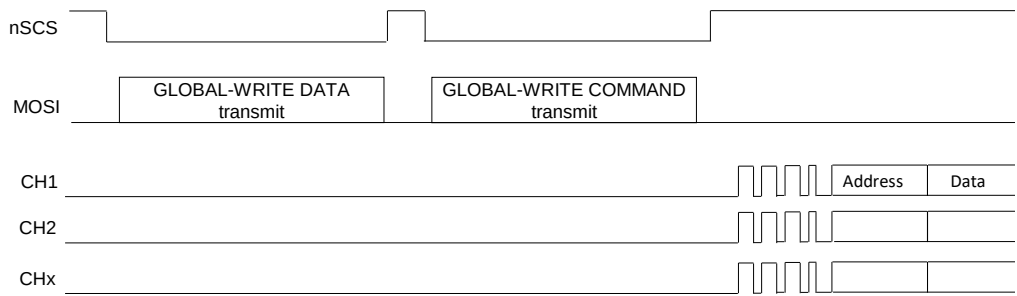


Fig. 13 The sequence of Global-write command

The GLOBAL-WRITE COMMAND register consists of 6 bits of ADDRESS field and 1 bit of START. The ADDRESS is the register address of the XD12 to be accessed. If the START bit is set with the ADDRESS together, the Global writing command begins issuing simultaneously in all activated channels, and the command streams repeat continuously for the number of daisied XD12. The number of daisied devices can be determined by the DAISY SIZE register of each channel. The max number of daisied devices per channel is 31.

Functional Description

Local-write

The Local-write command is used to write the register data to the XD12 in case that all XD12 devices daisied to a channel of the XC24 have the different register data each other. If the register data to be written to the XD12 are same, at that time the Global-write command is more useful for fast transaction time. For proper execution of the Local-write command, the LOCAL-RW DATA registers should be firstly written with desired value and then, the START bit of the LOCAL-WRITE COMMAND register should be set with desired address for beginning this command. It is careful that the LOCAL-RW POINTER RESET register must be set necessarily to 1 for the next transaction after the current Local - write command execution is completed or before the next Local-write command will start again.

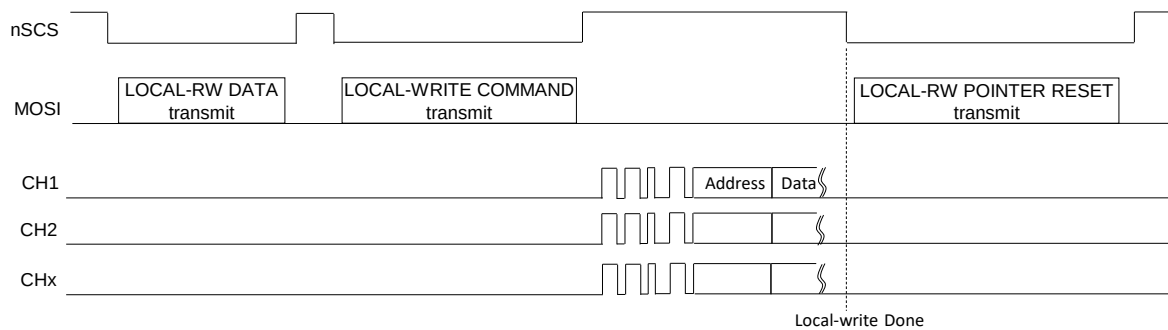


Fig. 14 The sequence of Local-write command

The LOCAL-WRITE COMMAND register consists of ADDRESS 6 bits, CH_SEG 2 bits and START 1 bit. The ADDRESS is the register address of the XD12 to be accessed. If the START bit is set with the ADDRESS together, the Local-write command begins the execution simultaneously in all activated channels, and the command streams repeat continuously for the number of daisied XD12. The number of daisied devices can be determined by the DAISY SIZE register of each channel. The max number of daisied devices per channel is 31. The channel segment bits, CH_SEG, are used to decide on which channels are activated for this command, because this command can be performed 8 channels at a time. The LOCAL-RW DATA registers are divided into a total of 8 port group and each port has a total of 16 registers. The first data register in the port is mapped to the first located XD12 in daisied order and the second data register is mapped to the second located XD12, and so on. Although the data registers in a port are a total 16, the XC-24 can support over 16 daisied devices in a channel by using the LOCAL-WR TRANSFER POINTER and LOCAL-RW DIFFERENCE POINTER. The LOCAL-WR TRANSFER DIFF POINTER of the LOCAL-RW DIFFERENCE POINTER shows how many data remain in the LOCAL-RW DATA buffers. A value of 0 means that the LOCAL-RW DATA buffers are empty, so you must continue to fill up before they are empty. A value greater than 16 means that the buffer has been overwritten, so you must wait until the pointer value is less than 16.

CH_SEG[1:0]	Channel to be activated	LOCAL-RW DATA 1 ~ 16							
		PORT1	PORT2	PORT3	PORT4	PORT5	PORT6	PORT7	PORT8
2'b00	Ch 1 ~ Ch 8	Ch 1	Ch 2	Ch 3	Ch 4	Ch 5	Ch 6	Ch 7	Ch 8
2'b01	Ch 9 ~ Ch 16	Ch 9	Ch 10	Ch 11	Ch 12	Ch 13	Ch 14	Ch 15	Ch 16
2'b10	Ch 17 ~ Ch 24	Ch 17	Ch 18	Ch 19	Ch 20	Ch 21	Ch 22	Ch 23	Ch 24

Table 8. The LOCAL-RW DATA mapping according to the CH_SEG

Functional Description

Local-RW DATA structure for access

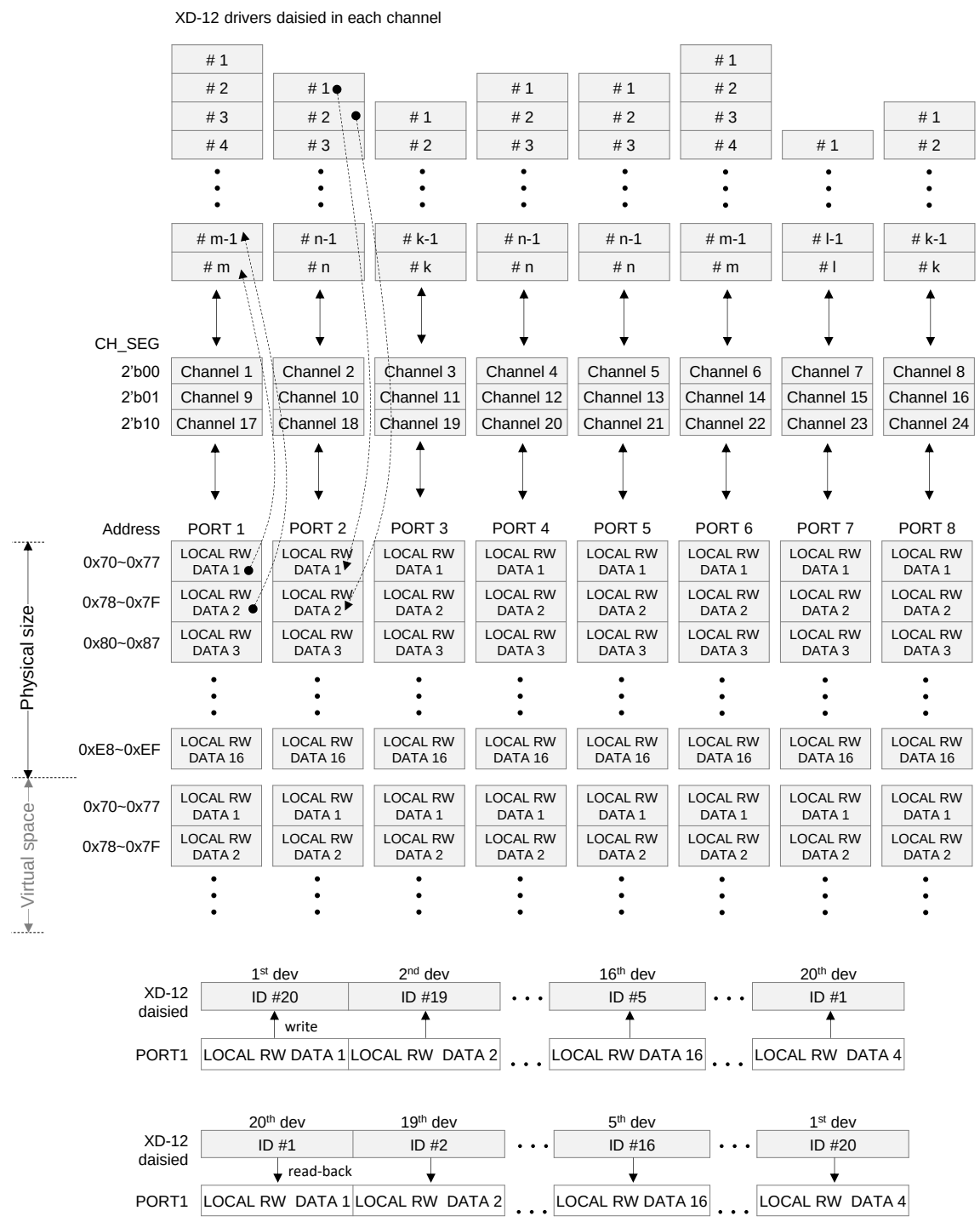


Fig. 15 LOCAL-RW DATA mapping structure and data flow

Functional Description

Sync-generation

The Sync-generation command is used to synchronize the frame sync time for all daisied devices simultaneously. This command execution can be done by two methods, the one is by setting the START bit of the SYNC-GEN COMMAND register and the other is by setting the SYNC_AUTO_EN bit of the COMMAND AUTO ENABLE register. If the SYNC_AUTO_EN bit is set, the Sync-generation command is automatically executed at the moment of receiving the VSYNC input. It is recommended to use this function, but it should be careful that The synchronization time of the XD12 devices may be as late as the number of XD12 devices than the time the VSYNC was received, so the VSYNC should arrive as soon as it is late. This command streams repeat continuously for the number of daisied devices in each channel.

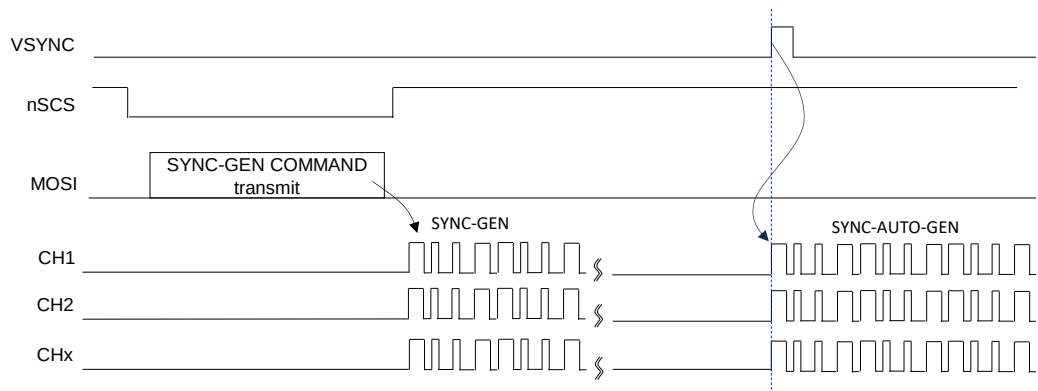


Fig. 16 The sequence of SYNC-GEN command

Fault-read

The Fault-read command is used to read-out the fault information of all daisied devices simultaneously. This command execution can be done by two methods, the one is by setting the START bit of the FAULT-GEN COMMAND register and the other is by setting the FAULT_AUTO_EN bit of the COMMAND AUTO ENABLE register. If the FAULT_AUTO_EN bit is set, the Fault-read command is automatically executed after the completion of the Sync auto-gen command at the moment of receiving the VSYNC input. The fault data to be read-back from all XD-12 devices in each channel is stored into the corresponding channel's GLOBAL FAULT-READ DATA register, respectively.

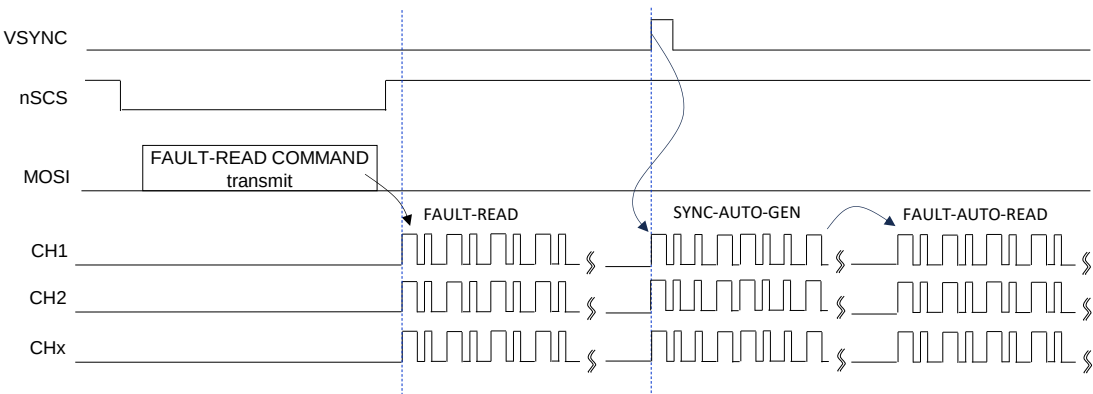


Fig. 17 The sequence of FAULT-READ command

Functional Description

Local-read

The Local-read command is used to read the register data from all XD12 devices daisied in a channel of the XC24. Unlike the Local-write or Global-write command, the Local-read command can be used to read the register data of the XD12 devices regardless of whether the data is the same or different. The data to be read-out from the XD12 devices are stored into the LOCAL-RW DATA registers sequentially like the Figure 10. The START bit of the LOCAL-READ COMMAND register should be set with desired address to be read-out for beginning this command. It is careful that the LOCAL-RW POINTER RESET register must be set necessarily to 1 for the next transaction after the current Local read command execution is completed or before the next Local-read command will start again.

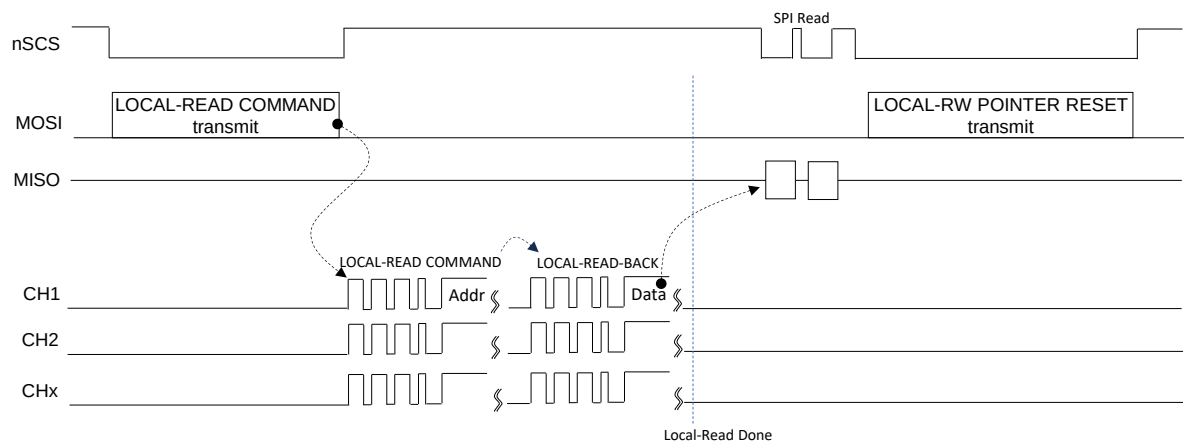


Fig. 18 The sequence of Local-read command

The LOCAL-READ COMMAND register consists of ADDRESS 6 bits, CH_SEG 2 bits and START 1 bit. The ADDRESS is the register address of the XD12 to be accessed. If the START bit is set with the ADDRESS together, the Local-read command begins the execution simultaneously in all activated channels, and the command streams repeat continuously for the number of daisied XD12. The number of daisied devices can be determined by the DAISY SIZE register of each channel. The max number of daisied devices per channel is 31. The channel segment bits, CH_SEG, are used to decide on which channels are activated for this command, because this command can be performed 8 channels at a time.

The LOCAL-RW DATA registers are divided into a total of 8 port group and each port has a total of 16 registers. Unlike the Local-write command, the first data register in the port is mapped to the last located XD-12 in daisied order and the second data register is mapped to the second located XD-12 from the back, and so on. Although the data registers in a port are a total 16, the XC-24 can support over 16 daisied devices in a channel by using the LOCAL-READ RECEIVE POINTER and LOCAL-RW DIFFERENCE POINTER. The LOCAL-RD RECEIVE DIFF POINTER of the LOCAL-RW DIFFERENCE POINTER shows how many data are filled up in the LOCAL-RW DATA buffers. A value of 0 means that the LOCAL-RW DATA buffers are empty, so you must wait for the buffers to be refilled, and if the value is greater than 16, it means that the LOCAL-RW DATA buffers have been overwritten, so you must drag out the buffers before they are overflowed. Refer to the Table 3. and the Figure 10. for the mapping scheme.

Functional Description

Sequence of Local Dimming data transfer

The Local Dimming data of the next frame should be transmitted to the XC24 through the SPI and the XC24 should complete to transfer the LD data to the XD12 devices before assertion of the next frame sync. And then, the LD data will be started dimming-out by the XC12 devices at the next frame sync. The LD_WR_POINTER of the LD WR POINTER register will be increased one by one in line unit whenever the LD data are written to the XC24 LD data buffers through the SPI and see the Figure 5. for the amount of one line LD. The LD_RD_POINTER will be increased one by one whenever the XC24 transmits the amount of one LD data to the XD12 devices. The LD_DIFF_POINTER is a value of the LD_WR_POINTER minus the LD_RD_POINTER, so the value is useful to check how many LD data remains in the internal LD buffers.

If the LD_TANS_START_POINTER value becomes equal to the LD_DIFF_POINTER, then the XC24 starts sending the LD data to the XD12 devices automatically until the LD_DIFF_POINTER value becomes to 0.

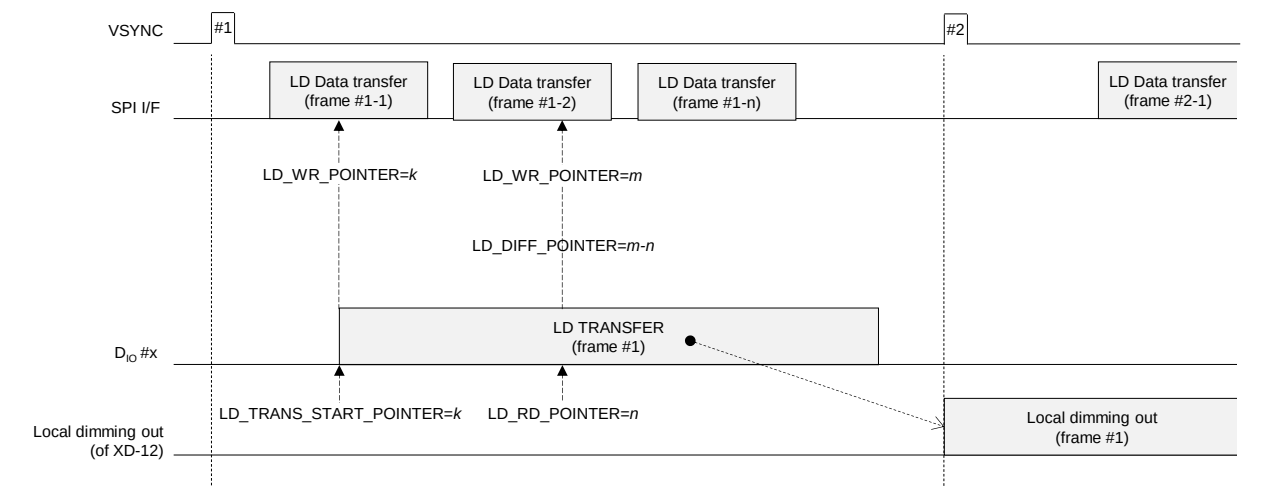


Fig. 19 The sequence of Local dimming Data transfer

Functional Description

Local Dimming data buffer

The total number of Local Dimming data buffers is 32 per channel and 1 buffer can contain one block data which consists of PAM and PWM data. The size of PAM and PWM data depends on the LD_WIDTH of CHANNEL_ENABLE_2 register and refer to previous chapter for more details. Even if the physical buffer size is limited to 32, the XC24 can support to transfer up to 511 blocks by using the LD_WRITE_POINTER, LD_READ_POINTER and LD_DIFFERENCE_POINTER. You can continue to send the LD data so that the LD_DIFFERENCE_POINTER does not become 0. The number of daisy devices of each channel can be different from each other by separately setting the DAISY_SIZE register of each channel. The below shows the scheme of LD buffers and the mapping to the daisy device. For more information refer to the example on the next page.

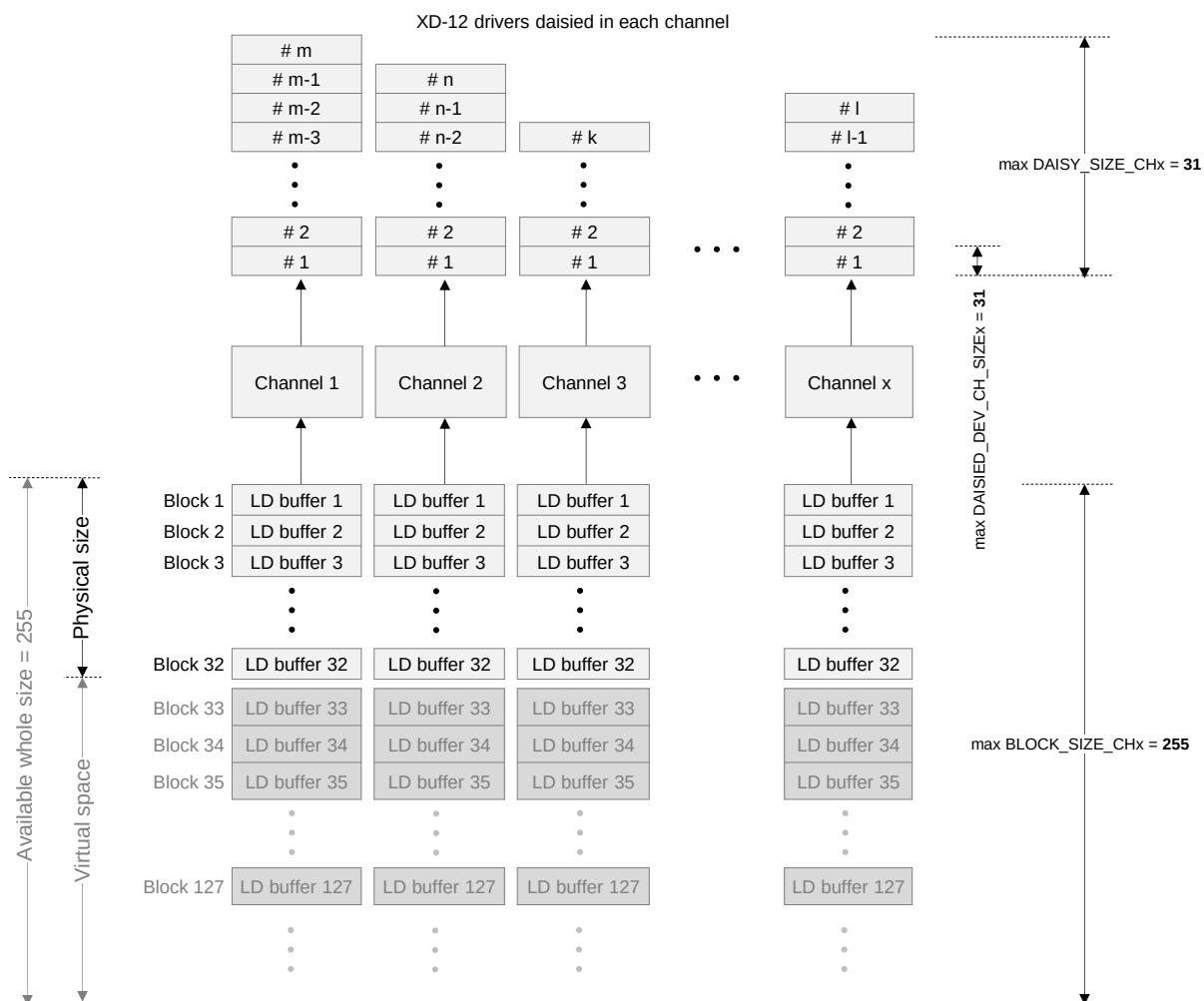


Fig. 20 The scheme of Local dimming Data buffers

Functional Description

Local Dimming data transfer

The below example shows how to set properly the registers relevant to the Local Dimming data transfer. The CH_SIZE of the CHANNEL ENABLE 2 register determines how many channels among a total of 24 channels of XC24 will be activated, and if 20 channels will be used, then the CH_SIZE is 0x14 and available channels become from the channel 1 to channel 20, so it should be careful that the order of channels to be selected must be a direction that sequentially increases from 1 to 24. The DAISY_SIZE_CHx of the DAISY SIZE registers determines how many XD12 devices will be daisy in the channel x, and if 10 XD12 devices are daisy in a channel x, then the DAISY_SIZE_CHx should be set to 0xA. The DAISIED_DEV_CH_SIZE determines how many channels of the XD12 will be used, so if the value is 0x4, the XD12 devices daisy in a channel x will capture only 4 LD data and then bypass the flowing LD data streams to the next XD12 devices and the remainder of the multiples of 4 be delivered lastly to a las XD12 device. It should be careful that all devices except for the last device must have the same value of DAISIED_DEV_CH_SIZE. The BLOCK_SIZE_CHx determines the total number of LD data to be transferred to a channel x.

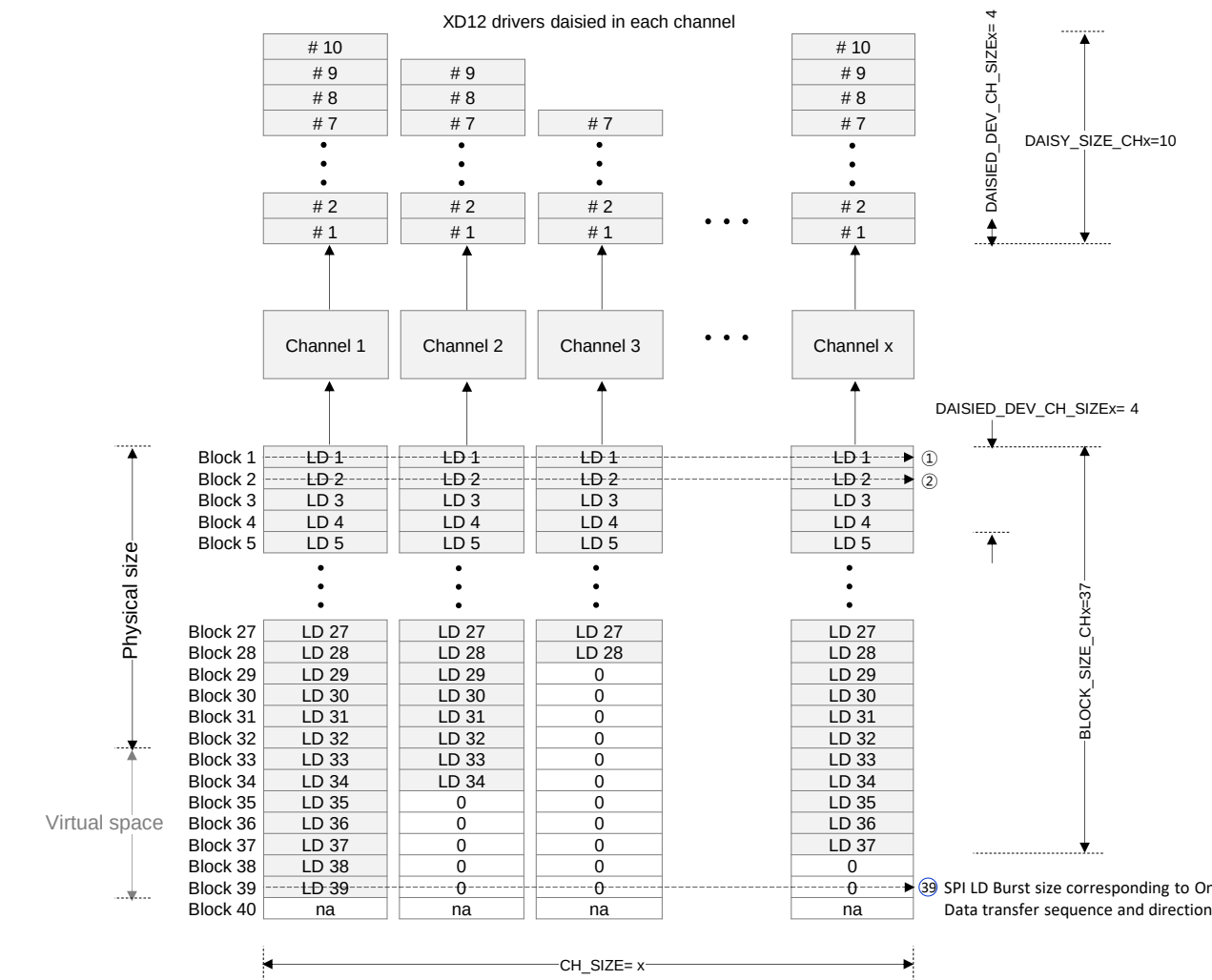


Fig. 21 The example of Local dimming Data-transfer

3. Register Map

The SPI interface protocol consists of two segments, the command 16-bit followed by the register's data 16-bit or a series of local dimming data.

Command packet format

B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
COMMAND CODE		START ADDRESS [7:0]								BURST SIZE [5:0]					

Bits	Fields	Descriptions
15:14	COMMAND CODE	2'b00 : No operation. 2'b01 : Register Read. 2'b10 : Register Write. 2'b11 : Local dimming data transfer.
13:6	START ADDRESS	The start address of registers to access.
5:0	BURST SIZE	Amount of data to be transmitted or received. In case of accessing the registers, the number means how many registers will be accessed for one time. In case of transmitting the LD data, the number means how many lines' LD data will be transmitted for one time, where one line means the number of enabled channels. If 20 channels are activated and the BURST SIZE is 10, and then the total amount of LD data to be transmitted becomes 20 * 10 * LD width/block [bit].

Data packet format

B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
DATA [15:0]															

Bits	Fields	Descriptions
15:0	DATA [15:0]	In case of COMMAND CODE : 2'b01 A register's data to read. 2'b10 A register's data to write. 2'b11 A series of the LD, local dimming data. Be careful that the data size and width to transfer the local dimming data are different from the register access. Refer to previous chapter 2. SPI protocol

SOFT RESET

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h00	0x0	x	x	x	x	x	x	x	x	x	x	VS_RST2	VS_RST1	x	RST3	RST2	RST1

Bits	Fields	Descriptions
5	VS_RST2	If this bit is set, all de-serializers' sm are reset at VSYNC rising. Normally, no need to use this bit.
4	VS_RST1	If this bit is set, all serializers' sm are reset at VSYNC rising. Normally, no need to use this bit.
2	RST3	Reset to all types of buffer, the frame buffers, the global and local r/w registers and the global fault read registers. Auto-cleared. Normally, no need to use this bit.
1	RST2	Reset to all serializers' and de-serializers' state machine including the timeout error flag, Auto-cleared. Normally, no need to use this bit.
0	RST1	Reset to all logics, Auto-cleared (No need to clear after set) It need to set this bit after POR to ensure XC-24 reset.

GLOBAL-WRITE COMMAND

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h01	0x0	START	x	x	x	x	x	x	x	x	x	ADDRESS [5:0]					

Bits	Fields	Descriptions
15	START	0 No operation. 1 Start executing the global-write command with the GLOBAL WRITE DATA register, together, and this bit is auto-cleared after completing this execution. For correct operation, It needs to set the GLOBAL WRITE DATA register with desired value before setting this bit. The global-writing can be used in case that the target registers' values of all XD-12 daisied devices are all the same. If the values are different from each other, then the local-write command should be used instead of the global-write command.
5:0	ADDRESS	The same target address of all XD-12 daisied devices to write.

LOCAL-WRITE COMMAND

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h02	0x0	START	x	x	x	x	x	CH_SEG [1:0]		x	x	ADDR [5:0]					

Bits	Fields	Descriptions
15	START	0 No operation. 1 Start executing the local-write command with the LOCAL RW DATA registers together, and this bit is auto-cleared after completing this execution. For correct operation, It needs to set the LOCAL RW DATA registers with desired values before setting this bit. The local-write command is used in case that the target registers' values of all XD-12 daisied devices are different from each other.
9:8	CH_SEG	These bits are used to designate which channels are activated for this command. 2'b00 The channel 1 from 8 are activated. 2'b01 The channel 9 from 16 are activated. 2'b10 The channel 17 from 24 are activated. 2'b11 No allocated. Refer to previous chapter. how to allocate the LOCAL RW DATA registers according to CH_SEG [1:0].
5:0	ADDRESS	The same target address of all XD-12 daisied devices to write

LOCAL-READ COMMAND

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h03	0x0	START	x	x	x	x	x	CH_SEG [1:0]		x	x	ADDR [5:0]					

Bits	Fields	Descriptions
15	START	0 No operation. 1 Start executing the local-read command with the LOCAL RW DATA registers together, and this bit is auto-cleared after completing this execution. For correct operation, It needs to set the LOCAL RW DATA registers with desired values before setting this bit. The local-read command is used to read out the target registers' values of all XD-12 daisied devices.
9:8	CH_SEG	These bits are used to designate which channels are activated for this command 2'b00 : The channel 1 from 8 are activated. 2'b01 : The channel 9 from 16 are activated. 2'b10 : The channel 17 from 24 are activated. 2'b11 : No allocated. Refer to previous chapter. how to allocate the LOCAL RW DATA registers according to CH_SEG.
5:0	ADDRESS	The same target address of all XD-12 daisied devices to read out.

ID-GEN COMMAND

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h04	0x0	START	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Bits	Fields	Descriptions	
15	START	0	No operation.
		1	Start executing the id-generating command to all XD-12 daisied devices and this bit is auto-cleared after completing this execution. All XD-12 daisied devices connected in each channel will have their own ID number after completing the id-generating command. This command is needed to read-out the XD-12 drivers' registers data correctly in sequential, so, this command should be executed before executing the local-reading or the fault-reading command. It is recommended that the id-generating command should be executed firstly on system initialization.

FAULT-READ COMMAND

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h05	0x0	START	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Bits	Fields	Descriptions	
15	START	0	No operation.
		1	Start executing the fault-reading command to all XD-12 daisied devices to monitor the fault event like as the thermal occurrence, the short status, the open status and the fb status of the XD-12 devices and this bit is auto-cleared after completing this execution. This command can be also auto-executed at the VSYNC rising edge if the FAULT_AUTO_EN bit in the COMMAND AUTO ENABLE register is set.

LD-TRANSFER COMMAND

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h06	0x0	START	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Bits	Fields	Descriptions	
15	START	0	No operation.
		1	Start executing the local dimming data transfer command to all XD-12 daisied devices in order to transfer the LD data in burst from the XC-24 to the XD-12 drivers and this bit is auto-cleared after the completion. This command can be also auto-executed immediately after the amount of LD data to be transferred to the XC-24 through the SPI interface is reached to the value of the LD_RD_START_POINTER of the LD START POINTER/TH register.

SYNC-GEN COMMAND

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h07	0x0	START	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Bits	Fields	Descriptions
15	START	0 No operation. 1 Start executing the sync-gen command to all XD12 daisied devices in order to make the frame synchronization to the drivers and this bit is auto-cleared after this execution. This command can be also auto-executed at the VSYNC rising edge, if the SYNC_ AUTO_EN bit in the COMMAND AUTO ENABLE register is set. It is recommended that the SYNC_AUTO_EN bit is set for good local dimming.

COMMAND AUTO ENABLE

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h08	0x111	x	x	x	x	x	x	x	TIMEOUT_EN	x	x	x	FAULT_AUTO_EN	x	x	x	SYNC_AUTO_EN

Bits	Fields	Descriptions
8	TIMEOUT_EN	0 No operation 1 The timeout function is activated during the execution of read-back commands. The timer starts from the end of completing whole command frame data out corresponding to all daisied device to first bit arrived time. The timer is expired at (MCLK * 8,192).
4	FAULT_AUTO_EN	0 No operation. 1 Start executing the fault-reading command automatically. If this bit is set, the fault-reading command is auto-executed in two conditions. The first is occurred at the time of VSYNC rising edge and the second condition is occurred at the time of expiration of the FAULT AUTO READ TIMER register for the number of occurrence of the FAULT AUTO READ EVENT register.
0	SYNC_AUTO_EN	0 No operation. 1 Start executing the sync-gen command automatically. If this bit is set, the sync-gen command is auto-executed at the time of VSYNC rising edge.

LD-WRITE POINTER

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h0A	RO	x	x	x	x	x	x	x	LD_WR_POINTER [8:0]								

Bits	Fields	Descriptions
8:0	LD_WR_POINTER	This register value represents the number of times the local dimming data are written into the internal LD buffers through the SPI interface. The number increases one by one whenever the local dimming data corresponding to (the number of enabled channels * one LD data width) are transferred to the internal LD buffers. This pointer value is reset at the VSYNC rising. LD_WIDTH of the CHANNEL ENABLE 2 register : 2'b00 : one block LD data width = 24 bits 2'b01 : one block LD data width = 24 bits 2'b10 : one block LD data width = 24 bits 2'b11 : one block LD data width = 16 bits

LD-READ POINTER

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h0B	RO	x	x	x	x	x	x	x									

LD_RD_POINTER [8:0]

Bits	Fields	Descriptions
8:0	LD_RD_POINTER	This register value represents the number of times the local dimming data are read-out to transfer the data from the internal LD buffers to the XD12 drivers. The number increases one by one whenever the local dimming data corresponding to the one block LD data of the XD12 drivers are read-out. This pointer value is reset at the VSYNC rising.

LD-DIFFERENCE POINTER

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h0C	RO	x	x	x	x	x	x	x	x	x	x						

LD_DIFF_POINTER [5:0]

Bits	Fields	Descriptions
5:0	LD_DIFF_POINTER	This value is useful to monitor how much the LD data remains in the internal LD buffers for checking overrun or underrun. = LD_WR_POINTER – LD_RD_POINTER

LD-TRANSFER START POINTER / THRESHOLD

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h0D	0x0	x	x														

LD_TRANS_START_POINTER [5:0]

INT_LD_SIGN

LD_DIFF_THRESHOLD [5:0]

Bits	Fields	Descriptions
13:8	LD_TRANS_START_POINTER	The LD_TRANS_START_POINTER determines when the transmit from the internal LD buffers to the XD12 drivers will begin. If the value is set to 8, then the transmit will begin as soon as the number of times the data through the SPI interface becomes 8. Once the transmit is stated, the transaction will be continued until the number of transmit reaches to the BLOCK_SIZE value of the BLOCK_SIZE registers.
7	INT_LD_SIGN	Don't set this bit.
5:0	LD_DIFF_THRESHOLD	The LD_DIFF_THRESHOLD is used to set in which condition the nINT_LD output signal will be issued to the host. Whenever the LD_DIFF_POINTER is equal to or smaller than the LD_DIFF_THRESHOLD, then the nINT_LD signal will be issued. This feature helps to check how much LD data remains in the internal LD buffer for monitoring overruns or underruns of the LD data.

LOCAL-WR TRANSFER POINTER

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h0E	RO	x	x	LOCAL_WR_OUT_POINTER [5:0]						x	x	LOCAL_WR_TRANS_POINTER [5:0]					

Bits	Fields	Descriptions
12:8	LOCAL_WR_OUT_POINTER	This register value represents the number of times the data from the LOCAL RW DATA registers are read-out to write the data to the XD12 drivers' registers. The number increases one by one whenever the data are read-out from the LOCAL RW DATA registers.
5:0	LOCAL_WR_TRANS_POINTER	This register value represents the number of times the data to be transferred to the XD12 drivers are written to the LOCAL RW DATA registers through the SPI interface. The number increased one by one whenever the data are written to the LOCAL RW DATA registers.

LOCAL-RD RECEIVE POINTER

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h0F	RO	x	x	LOCAL_RD_OUT_POINTER [5:0]						x	x	LOCAL_RD_REC_POINTER [5:0]					

Bits	Fields	Descriptions
12:8	LOCAL_RD_OUT_POINTER	This register value represents the number of times the data transferred from the XD12 drivers are read out from the LOCAL RW DATA registers for transferring the data through the SPI interface. The number increases one by one whenever the data are read-out from the LOCAL RW DATA registers.
5:0	LOCAL_RD_REC_POINTER	This register value represents the number of times the data to be transferred through the SPI interface are written from the XD-12 drivers to the LOCAL RW DATA registers. The number increased one by one whenever the data are written to the LOCAL RW DATA registers.

LOCAL-RW DIFFERENCE POINTER

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h10	RO	x	x	LOCAL_RD_DIFF_POINTER [5:0]						x	x	LOCAL_WR_DIFF_POINTER [5:0]					

Bits	Fields	Descriptions
12:8	LOCAL_RD_DIFF_POINTER	This value is useful to monitor how much the data from the XD12 drivers remains in the LOCAL RW DATA registers for checking overrun or underrun. = LOCAL_RD_REC_POINTER – LOCAL_RD_OUT_POINTER
5:0	LOCAL_WR_DIFF_POINTER	This value is useful to monitor how much the data to be transferred to the XD12 remains in the LOCAL RW DATA registers for checking overrun or underrun. = LOCAL_WR_TRANS_POINTER – LOCAL_WR_OUT_POINTER

LOCAL-RW POINTER RESET

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8		B7	B6	B5	B4	B3	B2	B1	B0	
'h11	0x0	x	x	x	x	x	x	x	LOCAL_RD_POINTER_RST		x	x	x	x	x	x	x	LOCAL_WR_POINTER_RST	

Bits Fields		Descriptions	
8	LOCAL_RD_POINTER_RST	0	No operation
		1	The LOCAL_RD_OUT_POINTER and the LOCAL_RD_REC_POINTER are reset. This bit is auto-cleared. After completing the Local read command, the relevant pointer value must be cleared by this bit before the next Local read command.
0	LOCAL_WR_POINTER_RST	0	No operation
		1	The LOCAL_WR_OUT_POINTER and the LOCAL_WR_TRANS_POINTER are reset. This bit is auto-cleared. After completing the Local write command, the relevant pointer value must be cleared by this bit before the next Local write command.

FAULT-AUTO-READ TIMER

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h12	0x0	FAULT_AUTO_RD_TIMER [15:0]															

Bits Fields		Descriptions	
15:0	FAULT_AUTO_RD_INTERVAL	This register determines the execution interval for the fault auto-read commands. Counted in MCLK units.	

FAULT-AUTO-READ EVENT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0	
'h13	0x0	FAULT_AUTO_RD_EVENT [7:0]								x	x	x	x	x	x	x	FAULT_AUTO_RD_INTERVAL [16]	

Bits Fields		Descriptions	
15:8	FAULT_AUTO_RD_EVENT	This register determines the execution times for the fault auto-read commands. The value is reset at VSYNC rising edge.	
0	FAULT_AUTO_RD_INTERVAL	The last bit of the FAULT_AUTO_RD_TIMER [16:0]	

SERIALIZER CLOCK GEN

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h14	0x0408	x	x	x	x	SCK_LOW [3:0]				x	x	x	SCK_HIGH [4:0]				

Bits	Fields	Descriptions
11:8	SCK_LOW	This register determines the Low pulse duration in case the serial data-out drives the low state. Counted in MCLK units.
4:0	SCK_HIGH	This register determines the high pulse duration in case the serial data-out drives the high state. Counted in MCLK units.

INTERRUPT ENABLE

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6
'h15	0x0	x	x	x	x	x	x	x	INT_TIMEOUT_ERR_EN	INT_FAULT_RD_FAIL_EN	INT_FAULT_AUTO_RD_FAIL_EN

Addr	Default	B5	B4	B3	B2	B1	B0
'h15	0x0	INT_RD_REC_FAIL_EN	INT_LD_EN	INT_THERMAL_EN	INT_SHORT_EN	INT_OPEN_EN	INT_FB_EN

Bits	Fields	Descriptions
8	INT_TIMEOUT_ERR_EN	1 This bit is able to issue the interrupt through the nINT_FAULT pin in case that the timer will be expired during the execution for read-back commands. The expired time is mclk * 8,192.
7	INT_FAULT_RD_FAIL_EN	1 This bit is able to issue the interrupt through the nINT_FAULT pin in case that any violation happens during the execution for the fault read-back from the XD12.
6	INT_FAULT_AUTO_RD_FAIL_EN	1 This bit is able to issue the interrupt through the nINT_FAULT pin in case that any violation happens during the execution for the fault auto-read-back from the XD12.
5	INT_RD_REC_FAIL_EN	1 This bit is able to issue the interrupt through the nINT_FAULT pin in case that any violation happens during the execution for the local read-back from the XD12.
4	INT_LD_EN	1 This bit is able to issue the interrupt through the nINT_FAULT pin in case that the LD_DIFF_POINTER is equal to or smaller than the LD_DIFF_THRESOLD during the execution of the LD transfer to the XD12.
3	INT_THERMAL_EN	1 This bit is able to issue the interrupt through the nINT_FAULT pin in case that the thermal regulation happens among the XD12 drivers during dimming operation.
2	INT_SHORT_EN	1 This bit is able to issue the interrupt through the nINT_FAULT pin in case that the short condition happens among the XD12 drivers during dimming operation.
1	INT_OPEN_EN	1 This bit is able to issue the interrupt through the nINT_FAULT pin in case that the open condition happens among the XD12 drivers during dimming operation.
0	INT_FB_EN	1 This bit is able to issue the interrupt through the nINT_FAULT pin in case that the fb condition happens among the XD12 drivers during dimming operation.

COMMAND STATUS 1

Addr	Default	B15	B14	B13	B12
'h16	RO	LOCAL_RD_DONE	LOCAL_RD_DOING	LOCAL_WR_DONE	LOCAL_WR_DOING

Addr	Default	B11	B10	B9	B8	B7	B6
'h16	RO	GLOBAL_WR_DONE	GOLBAL_WR_DOING	LD_TRANS_DONE	LD_TRANS_DOING	FAULT_DONE	FAULT_DOING

Addr	Default	B5	B4	B3	B2	B1	B0
'h16	RO	FAULT_AUTO_DONE	FAULT_AUTO_DOING	SYNC_AUTO_DONE	SYNC_AUTO_DOING	SYNC_DONE	SYNC_DOING

Bits	Fields	Descriptions
15	LOCAL_RD_DONE	This bit is set after the execution for the local-reading command is completed.
14	LOCAL_RD_DOING	This bit is set while the local-reading command is working until it is complete.
13	LOCAL_WR_DONE	This bit is set after the execution for the local-writing command is completed.
12	LOCAL_WR_DOING	This bit is set while the local-writing command is working until it is complete.
11	GLOBAL_WR_DONE	This bit is set after the execution for the global-writing command is completed.
10	GLOBAL_WR_DOING	This bit is set while the global-writing command is working until it is complete.
9	LD_TRANS_DONE	This bit is set after the execution for the LD transferring command is completed.
8	LD_TRANS_DOING	This bit is set while the LD transferring command is working until it is complete.
7	FAULT_DONE	This bit is set after the execution for the fault-reading command is completed.
6	FAULT_DOING	This bit is set while the fault-reading command is working until it is complete.
5	FAULT_AUTO_DONE	This bit is set after the execution for the fault auto-reading command is completed.
4	FAULT_AUTO_DOING	This bit is set while the fault auto-reading command is working until it is complete.
3	SYNC_AUTO_DONE	This bit is set after the execution for the sync auto-gen command is completed.
2	SYNC_AUTO_DOING	This bit is set while the sync auto-gen command is working until it is complete.
1	SYNC_DONE	This bit is set after the execution for the sync-gen command is completed.
0	SYNC_DOING	This bit is set while the sync-gen command is working until it is complete.

COMMAND STATUS 2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h17	RO	x	x	x	x	x	x	x	x	x	x	x	x	x	x	ID_GEN_DONE	ID_GEN_DOING

Bits	Fields	Descriptions
1	ID_GEN_DONE	This bit is set after the execution for the id-gen command is completed.
0	ID_GEN_DOING	This bit is set while the id-gen command is working until it is complete.

RECEIVE STATUS

Addr	Default	B15	B14	B13	B12
'h18	RO	x	FAULT_REC_FAIL	FAULT_AUTO_REC_FAIL	LOCAL_REC_FAIL

Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4
'h18	RO	x	x	x	TIMEOUT_ERR	x	x	FAULT_REC_DONE	FAULT_REC_DOING

Addr	Default	B3	B2	B1	B0
'h18	RO	FAULT_AUTO_REC_DONE	FAULT_AUTO_REC_DOING	LOCAL_REC_DONE	LOCAL_REC_DOING

Bits	Fields	Descriptions
14	FAULT_REC_FAIL	This bit is set if any violation happens while the fault-receiving job is working.
13	FAULT_AUTO_REC_FAIL	This bit is set if any violation happens while the fault auto-receiving job is working.
12	LOCAL_REC_FAIL	This bit is set if any violation happens while the local read-receiving job is working.
8	TIMEOUT_ERR	This bit is set if the timer will be expired while the execution of read-back due to no response from the XD devices. This bit is auto-cleared after the readout of the INTERRUPT STATUS register.
5	FAULT_REC_DONE	This bit is set after the fault-receiving job is completed.
4	FAULT_REC_DOING	This bit is set while the fault-receiving job is working until it is complete.
3	FAULT_AUTO_REC_DONE	This bit is set after the fault auto-receiving job is completed.
2	FAULT_AUTO_REC_DOING	This bit is set while the fault auto-receiving job is working until it is complete.
1	LOCAL_REC_DONE	This bit is set after the local read-receiving job is completed.
0	LOCAL_REC_DOING	This bit is set while the local read-receiving job is working until it is complete.

INTERRUPT STATUS

Addr	Default	B15	B14	B13	B12	B11	B10	B9		
'h19	RO	x	x	x	INT_TIMEOUT_ERR_SRC	INT_FAULT_REC_FAIL_SRC	INT_FAULT_AUTO_REC_FAIL_SRC	INT_LOCAL_REC_FAIL_SRC		

Addr	Default	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h19	RO	INT_LD_TRAN_SRC	INT_TML_SRC	INT_SHORT_SRC	INT_OPEN_SRC	INT_FB_SRC	x	x	INT_LD	INT_FAULT

Bits	Fields	Descriptions
12	INT_TIMEOUT_ERR_SRC	This bit shows whether the timeout has been happened while the read-back commands are working. This bit is auto-cleared after the readout of INTERRUPT STATUS register.
11	INT_FAULT_REC_FAIL_SRC	This bit shows whether any violation has been happened while the fault-receiving job is working. This bit value will remain at the same level until start of the next command even if this register has been read-out.
10	INT_FAULT_AUTO_REC_FAIL_SRC	This bit shows whether any violation has been happened while the fault auto-receiving job is working. This bit value will remain at the same level until start of the next command even if this register has been read-out.
9	INT_LOCAL_REC_FAIL_SRC	This bit shows whether any violation has been happened while the local read-receiving job is working. This bit value will remain at the same level until start of the next command even if this register has been read-out.
8	INT_LD_TRANS_SRC	This bit shows whether the LD interrupt has been asserted while transferring the LD data to XD12 drivers through the XC-24 controller. This bit value will remain at the same level until start of the next command even if this register has been read-out.
7	INT_TML_SRC	This bit shows the THERMAL status from all XD12 daisied to active channels after the FAULT-(AUTO)-READ execution. This bit value will be kept the current level until newly updated from a next FAULT-(AUTO)-READ execution even if this register has been read-out.
6	INT_SHORT_SRC	This bit shows the SHORT status from all XD12 daisied to active channels after the FAULT-(AUTO)-READ execution. This bit value will be kept the current level until newly updated from a next FAULT-(AUTO)-READ execution even if this register has been read-out.
5	INT_OPEN_SRC	This bit shows the OPEN status from all XD12 daisied to active channels after the FAULT-(AUTO)-READ execution. This bit value will be kept the current level until newly updated from a next FAULT-(AUTO)-READ execution even if this register has been read-out.
4	INT_FB_SRC	This bit shows the FB status from all XD12 daisied to the active channels after the FAULT-(AUTO)-READ execution. This bit value will be kept the current level until newly updated from a next FAULT-(AUTO)-READ execution even if this register has been read-out.
1	INT_LD	If this bit is set, the nINT_LD pin will be asserted to the active-low. This bit represents an interrupt sources, INT_LD_TRAN_SRC if its interrupt enable bits is set. If the INT_LD_EN bit is set and INT_LD_TRAN_SRC is asserted, then, this bit will be asserted through nINT_LD pin. This bit is auto-cleared after this INTERRUPT STATUS register is read-out.
0	INT_FAULT	If this bit is set, the nINT_FAULT pin will be asserted to the active-low. This bit represents several interrupt sources, INT_FB_SRC, INT_OPEN_SRC, INT_SHORT_SRC, INT_TML_SRC, INT_LOCAL_REC_FAIL_SRC, INT_FAULT_AUTO_REC_FAIL_SRC and INT_FAULT_REC_FAIL_SRC if their own interrupt enable bits are set. If the corresponding interrupt enable bit is set and the interrupt source is asserted, then, this bit will be asserted through nINT_FAULT pin. This bit is auto-cleared after this INTERRUPT STATUS register is read-out.

FB PWM PERIOD

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h1A	0x0	FB_PWM_EN	FB_PWM_POL	T_FB_EN	x	x	FB_PWM_PERIOD [10:0]										

Bits	Fields	Descriptions
15	FB_PWM_EN	The FB signal, the feedback to adjust the VLED, can be driven in DC level or PWM duty on the external FB pin, the open-drain. 0 The FB signal is driven in DC level on the FB pin, and the level will be newly updated from the FAULT_CHx[0] of the GLOBAL FAULT-READ DATA registers whenever the FAULT-AUTO-READ commands have been executed. The FB signal is made from ORing all FAULT_CHx[0] of active channels. 1 The FB signal is driven in PWM duty on the FB pin, not the DC level. The PWM signal is determined by the FB_PWM_PERIOD, the FB_PWM_DUTY_INC, the FB_PWM_DUTY_DEC and FB_PWM_DUTY_WAIT. The FB signal in PWM mode will be also updated at every FAULT-(AUTO)-READ execution.
14	FB_PWM_POL	This bit determines a polarity of the FB PWM signal toward the external FB pin. If this bit '0', the FB assertion will be showed in a low level on the FB pin because the pin has an open-drain property.
13	T_FB_EN	If this bit '1', the FB signal toward the FB pin will be de-asserted as long as the thermal event is active, even though the FB event happens.
10:0	FB_PWM_PERIOD	This register value determines the period of PWM and counted in a MCLK. The 20KHz is recommended, but the value depends on the system condition for externa FB control circuit.

FB PWM DUTY 1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h1B	0x0	x	x	x	x	x	x	x	FB_PWM_DUTY_INC [8:0]								

Bits	Fields	Descriptions
8:0	FB_PWM_DUTY_INC	This register determines the amount of increment to the FB_PWM_DUTY. As soon as the FB event is sensed after the FAULT-(AUTO)-READ execution, the FB_PWM_DUTY will be increased as many as the FB_PWM_DUTY_INC value. If the FB_PWM_DUTY is reached to 100%, then the value will be not increased any more. If the value is changed during working, the value will be immediately applied to the FB_PWM_DUTY. For details, refer to the Fig 21.

FB PWM DUTY 2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h1C	0x0	FB_PWM_DUTY_WAIT [3:0]				x	x	x	x	FB_PWM_DUTY_DEC [7:0]							

Bits	Fields	Descriptions
15:12	FB_PWM_DUTY_WAIT	This determines how many waiting-times of the FAULT-(AUTO)-READ execution the FB event does not occur successively, and if the waiting-times is expired, then the FB_PWM_DUTY will be reduced as many as the FB_PWM_DUTY_DEC value. If the FB_PWM_DUTY is reached to the FB_PWM_DUTY_BASE, the value will be not decreased any more even if the waiting-times are expired. The minimum value, 1, means the FB_PWM_DUTY will be decreased at every FAULT-(AUTO)-READ execution if it results in no FAULT_CHx[1]. For details, refer to the Fig 21.
7:0	FB_PWM_DUTY_DEC	This register determines the amount of decrement from the FB_PWM_DUTY. As soon as the FB event will be sensed after the FAULT_AUTO_READ execution, the FB_PWM_DUTY will be decreased as many as the FB_PWM_DUTY_DEC value. If the FB_PWM_DUTY is reached to 0, then the value will be not decreased any more. If the value is changed during working, the value will be immediately applied to the FB_PWM_DUTY. For details, refer to the Fig 21.

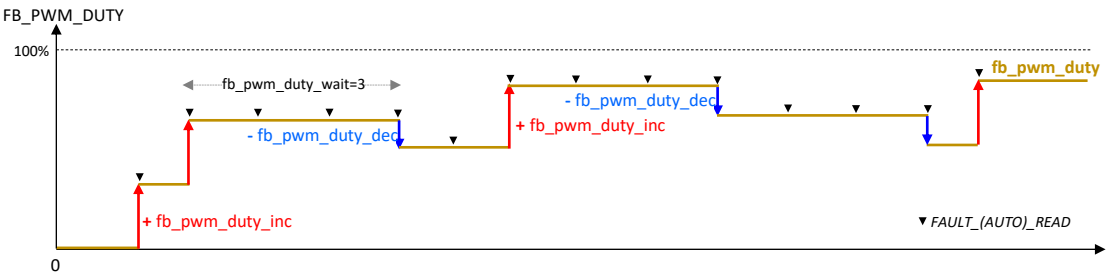


Fig 21. The behavior of FB PWM DUTY

FB PWM DUTY 3

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h1D	RO	x	x	x	x	x	FB_PWM_DUTY [10:0]										

Bits	Fields	Descriptions
10:0	FB_PWM_DUTY	This register shows the PWM duty-high value being currently work. For details, refer to the Fig 21.

COMMAND LATENCY

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h1F	0x1F	x	x	x	x	x	x	x	x	CMD_LATENCY [7:0]							

Bits	Fields	Descriptions
7:0	CMD_LATENCY	This determines the interval time between the end of a command and the start of a new command. The value can be increased in the read command execution if the XD12 mclk is so slower than the XC24 mclk, but it is recommended to keep the default value in most cases. The max value which is possible to set is decimal 99.

DAISIED DEVICE CHANNEL SIZE 1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h20	0x0	x	DAISIED_DEV_CH_SIZE3 [4:0]					DAISIED_DEV_CH_SIZE2 [4:0]					DAISIED_DEV_CH_SIZE1 [4:0]				

DAISIED DEVICE CHANNEL SIZE 2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h21	0x0	x	DAISIED_DEV_CH_SIZE6 [4:0]					DAISIED_DEV_CH_SIZE5 [4:0]					DAISIED_DEV_CH_SIZE4 [4:0]				

DAISIED DEVICE CHANNEL SIZE 3

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h22	0x0	x	DAISIED_DEV_CH_SIZE9 [4:0]					DAISIED_DEV_CH_SIZE8 [4:0]					DAISIED_DEV_CH_SIZE7 [4:0]				

DAISIED DEVICE CHANNEL SIZE 4

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h23	0x0	x	DAISIED_DEV_CH_SIZE12 [4:0]					DAISIED_DEV_CH_SIZE11 [4:0]					DAISIED_DEV_CH_SIZE10 [4:0]				

DAISIED DEVICE CHANNEL SIZE 5

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h24	0x0	x	DAISIED_DEV_CH_SIZE15 [4:0]					DAISIED_DEV_CH_SIZE14 [4:0]					DAISIED_DEV_CH_SIZE13 [4:0]				

DAISIED DEVICE CHANNEL SIZE 6

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h25	0x0	x	DAISIED_DEV_CH_SIZE18 [4:0]					DAISIED_DEV_CH_SIZE17 [4:0]					DAISIED_DEV_CH_SIZE16 [4:0]				

DAISIED DEVICE CHANNEL SIZE 7

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h26	0x0	x	DAISIED_DEV_CH_SIZE218 [4:0]					DAISIED_DEV_CH_SIZE20 [4:0]					DAISIED_DEV_CH_SIZE19 [4:0]				

DAISIED DEVICE CHANNEL SIZE 8

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h27	0x0	x	DAISIED_DEV_CH_SIZE24 [4:0]					DAISIED_DEV_CH_SIZE23 [4:0]					DAISIED_DEV_CH_SIZE22 [4:0]				

Bits	Fields	Descriptions
14:10 9:5 4:0	DAISY_DEV_ CH_SIZEx	This register determines how many channels of the XD12 daisied are activated. Every XD12 daisied must have the same number of active channels but the last device in daisy can have the different number of active channels. All DAISIED_DEV_CH_SIZEx registers' value must be same, but currently even if the DAISIED_DEV_CH_SIZE1 is only set, the value will have the same effect on all other DAISIED_DEV_CH_SIZE 2 ~ 24.

LDO

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h2A	0x0008	x	x	x	x	x	x	x	x	x	x	x	x		LDO [3:0]		

Bits	Fields	Descriptions
3:0	LDO	This is for adjusting the internal LDO and it is recommended to use the default value.

CONFIDENTIAL

DAISY SIZE 1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h30	0x0	x	DAISY_SIZE_CH3 [4:0]					DAISY_SIZE_CH2 [4:0]					DAISY_SIZE_CH1 [4:0]				

DAISY SIZE 2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h31	0x0	x	DAISY_SIZE_CH6 [4:0]					DAISY_SIZE_CH5 [4:0]					DAISY_SIZE_CH4 [4:0]				

DAISY SIZE 3

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h32	0x0	x	DAISY_SIZE_CH9 [4:0]					DAISY_SIZE_CH8 [4:0]					DAISY_SIZE_CH7 [4:0]				

DAISY SIZE 4

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h33	0x0	x	DAISY_SIZE_CH12 [4:0]					DAISY_SIZE_CH11 [4:0]					DAISY_SIZE_CH10 [4:0]				

DAISY SIZE 5

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h34	0x0	x	DAISY_SIZE_CH15 [4:0]					DAISY_SIZE_CH14 [4:0]					DAISY_SIZE_CH13 [4:0]				

DAISY SIZE 6

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h35	0x0	x	DAISY_SIZE_CH18 [4:0]					DAISY_SIZE_CH17 [4:0]					DAISY_SIZE_CH16 [4:0]				

DAISY SIZE 7

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h36	0x0	x	DAISY_SIZE_CH21 [4:0]					DAISY_SIZE_CH20 [4:0]					DAISY_SIZE_CH19 [4:0]				

DAISY SIZE 8

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h37	0x0	x	DAISY_SIZE_CH24 [4:0]					DAISY_SIZE_CH23 [4:0]					DAISY_SIZE_CH22 [4:0]				

Bits	Fields	Descriptions
14:10 9:5 4:0	DAISY_SIZE_CHx	This register should be set with a valid number for correct operating. The number means how many XD12 drivers are daisied in the corresponding channel. The maximum 31 XD12 drivers per channel can be daisied.

BLOCK SIZE 1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h38	0x0	BLOCK_SIZE_CH2 [7:0]								BLOCK_SIZE_CH1 [7:0]							

BLOCK SIZE 2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h39	0x0	BLOCK_SIZE_CH4 [7:0]								BLOCK_SIZE_CH3 [7:0]							

BLOCK SIZE 3

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h3A	0x0	BLOCK_SIZE_CH6 [7:0]								BLOCK_SIZE_CH5 [7:0]							

BLOCK SIZE 4

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h3B	0x0	BLOCK_SIZE_CH8 [7:0]								BLOCK_SIZE_CH7 [7:0]							

BLOCK SIZE 5

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h3C	0x0	BLOCK_SIZE_CH10 [7:0]								BLOCK_SIZE_CH9 [7:0]							

BLOCK SIZE 6

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h3D	0x0	BLOCK_SIZE_CH12 [7:0]								BLOCK_SIZE_CH11 [7:0]							

BLOCK SIZE 7

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h3E	0x0	BLOCK_SIZE_CH14 [7:0]								BLOCK_SIZE_CH13 [7:0]							

BLOCK SIZE 8

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h3F	0x0	BLOCK_SIZE_CH16 [7:0]								BLOCK_SIZE_CH15 [7:0]							

BLOCK SIZE 9

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h40	0x0	BLOCK_SIZE_CH18 [7:0]								BLOCK_SIZE_CH17 [7:0]							

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BLOCK SIZE 10

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h41	0x0	BLOCK_SIZE_CH20 [7:0]								BLOCK_SIZE_CH19 [7:0]							

BLOCK SIZE 11

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h42	0x0	BLOCK_SIZE_CH22 [7:0]								BLOCK_SIZE_CH21 [7:0]							

BLOCK SIZE 12

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h43	0x0	BLOCK_SIZE_CH24 [7:0]								BLOCK_SIZE_CH23 [7:0]							

Bits	Fields	Descriptions
15:8 7:0	BLOCK_SIZE_CHx	This register should be set with a valid number for correct operating. The number means how many block zones are daisied in the corresponding channel. The maximum 255 block zones per channel can be daisied.

CHANNEL ENABLE 1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8
'h45	0x0	CH16_EN	CH15_EN	CH14_EN	CH13_EN	CH12_EN	CH11_EN	CH10_EN	CH9_EN

Addr	Default	B7	B6	B5	B4	B3	B2	B1	B0
'h45	0x0	CH8_EN	CH7_EN	CH6_EN	CH5_EN	CH4_EN	CH3_EN	CH2_EN	CH1_EN

Bits	Fields	Descriptions
15:0	CHx_EN	These bits should be set if the corresponding channels are to be used, and don't set if not used. It is careful that <i>these bits should be set consecutively</i> , such as CH1_EN, CH2_EN, CH3_EN, and so on, and the number of enabled channels does not need to match the CH_SIZE value.

CHANNEL ENABLE 2/ SIZE/ LD WIDTH

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8
'h46	0x0	LD_WIDTH [1:0]		x	CH_SIZE [4:0]				

Addr	Default	B7	B6	B5	B4	B3	B2	B1	B0
'h46	0x0	CH24_EN	CH23_EN	CH22_EN	CH21_EN	CH20_EN	CH19_EN	CH18_EN	CH17_EN

Bits	Fields	Descriptions
15:14	LD_WIDTH	These bits represent the number of bits contained in one LD data in one block zone. 2'b00 : 24 bits/block (even if the 24 bits/block are transferred in SPI interface, the valid bits are PAM 8 bits and PWM 14 bits) 2'b01 : 24 bits/block, PAM 10 bits + PWM 11 bits. 2'b10 : 24 bits/block, PAM 10 bits + PWM 12 bits. 2'b11 : 16 bits/block
12:8	CH_SIZE	This value determines the amount of LD data to be transmitted through the SPI interface for one time. This value is not needed to match with the number of enabled channels in the CHANNEL ENABLE register. The LD data to be transmitted will be mapped sequentially from channel 1 to channel n regardless of the corresponded CHx_EN bit activation.
7:0	CHx_EN	The same as above CHANNEL ENABLE 1 register.

GLOBAL-WRITE DATA

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h60	0x0	x	x	x	x	GLOBAL_WR_DATA [11:0]											

Bits	Fields	Descriptions
11:0	GLOBAL_WR_DATA	This register represents the data to be written to the XD12 drivers daisied. This register should be written with desired value before the global-writing command is executed.

GLOBAL FAULT-READ DATA 1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h61	0x0	FAULT_CH4 [3:0]				FAULT_CH3 [3:0]				FAULT_CH2 [3:0]				FAULT_CH1 [3:0]			

GLOBAL FAULT-READ DATA 2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h62	0x0	FAULT_CH8 [3:0]				FAULT_CH7 [3:0]				FAULT_CH6 [3:0]				FAULT_CH5 [3:0]			

GLOBAL FAULT-READ DATA 3

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h63	0x0	FAULT_CH12 [3:0]				FAULT_CH11 [3:0]				FAULT_CH10 [3:0]				FAULT_CH9 [3:0]			

GLOBAL FAULT-READ DATA 4

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h64	0x0	FAULT_CH16 [3:0]				FAULT_CH15 [3:0]				FAULT_CH14 [3:0]				FAULT_CH13 [3:0]			

GLOBAL FAULT-READ DATA 5

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h65	0x0	FAULT_CH20 [3:0]				FAULT_CH19 [3:0]				FAULT_CH18 [3:0]				FAULT_CH17 [3:0]			

GLOBAL FAULT-READ DATA 6

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h66	0x0	FAULT_CH24 [3:0]				FAULT_CH23 [3:0]				FAULT_CH22 [3:0]				FAULT_CH21 [3:0]			

Bits	Fields	Descriptions
15:12 11:8 7:4 3:0	FAULT_CHx	These bits represent the fault status from all XD12 drivers daisied in each channel. FAULT_CHx [3] : The THERMAL status, all bits from each channel are OR-ed. FAULT_CHx [2] : The SHORT status, all bits from each channel are OR-ed. FAULT_CHx [1] : The OPEN status, all bits from each channel are OR-ed. FAULT_CHx [0] : The FB status, all bits from each channel are OR-ed. The value will be kept the current state until the value is newly updated from a next FAULT-(AUTO)-READ execution. The OR-ed signal of the FAULT_CHx [0] is driven directly to the FB pin and the pattern depends on the FB_PWM_EN bit.

PORT1 LOCAL-RW DATA 1 ~ PORT8 LOCAL RW DATA 1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h70~'h77	0x0	x	x	x	x	PORT1_LOCAL_RW_DATA 1 [11:0] ~ PORT8_LOCAL_RW_DATA 1 [11:0]											

PORT1 LOCAL-RW DATA 2 ~ PORT8 LOCAL RW DATA 2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h78~'h7F	0x0	x	x	x	x	PORT1_LOCAL_RW_DATA 2 [11:0] ~ PORT8_LOCAL_RW_DATA 2 [11:0]											

PORT1 LOCAL-RW DATA 3 ~ PORT8 LOCAL RW DATA 3

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h80~'h87	0x0	x	x	x	x	PORT1_LOCAL_RW_DATA 3 [11:0] ~ PORT8_LOCAL_RW_DATA 3 [11:0]											

PORT1 LOCAL-RW DATA 4 ~ PORT8 LOCAL RW DATA 4

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h88~'h8F	0x0	x	x	x	x	PORT1_LOCAL_RW_DATA 4 [11:0] ~ PORT8_LOCAL_RW_DATA 4 [11:0]											

PORT1 LOCAL-RW DATA 5 ~ PORT8 LOCAL RW DATA 5

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h90~'h97	0x0	x	x	x	x	PORT1_LOCAL_RW_DATA 5 [11:0] ~ PORT8_LOCAL_RW_DATA 5 [11:0]											

PORT1 LOCAL-RW DATA 6 ~ PORT8 LOCAL RW DATA 6

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h98~'h9F	0x0	x	x	x	x	PORT1_LOCAL_RW_DATA 6 [11:0] ~ PORT8_LOCAL_RW_DATA 6 [11:0]											

PORT1 LOCAL-RW DATA 7 ~ PORT8 LOCAL RW DATA 7

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hA0~'hA7	0x0	x	x	x	x	PORT1_LOCAL_RW_DATA 7 [11:0] ~ PORT8_LOCAL_RW_DATA 7 [11:0]											

PORT1 LOCAL-RW DATA 8 ~ PORT8 LOCAL RW DATA 8

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hA8~'hAF	0x0	x	x	x	x	PORT1_LOCAL_RW_DATA 8 [11:0] ~ PORT8_LOCAL_RW_DATA 8 [11:0]											

PORT1 LOCAL-RW DATA 9 ~ PORT8 LOCAL RW DATA 9

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hB0~'hB7	0x0	x	x	x	x	PORT1_LOCAL_RW_DATA 9 [11:0] ~ PORT8_LOCAL_RW_DATA 9 [11:0]											

PORT1 LOCAL-RW DATA 10 ~ PORT8 LOCAL RW DATA 10

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hB8~'hBF	0x0	x	x	x	x	PORT1_LOCAL_RW_DATA 10 [11:0] ~ PORT8_LOCAL_RW_DATA 10 [11:0]											

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PORT1 LOCAL-RW DATA 11 ~ PORT8 LOCAL RW DATA 11

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hC0~'hC7	0x0	x	x	x	x	PORT1_LOCAL_RW_DATA 11 [11:0] ~ PORT8_LOCAL_RW_DATA 11 [11:0]											

PORT1 LOCAL-RW DATA 12 ~ PORT8 LOCAL RW DATA 12

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hC8~'hCF	0x0	x	x	x	x	PORT1_LOCAL_RW_DATA 12 [11:0] ~ PORT8_LOCAL_RW_DATA 12 [11:0]											

PORT1 LOCAL-RW DATA 13 ~ PORT8 LOCAL RW DATA 13

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hD0~'hD7	0x0	x	x	x	x	PORT1_LOCAL_RW_DATA 13 [11:0] ~ PORT8_LOCAL_RW_DATA 13 [11:0]											

PORT1 LOCAL-RW DATA 14 ~ PORT8 LOCAL RW DATA 14

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hD8~'hDF	0x0	x	x	x	x	PORT1_LOCAL_RW_DATA 14 [11:0] ~ PORT8_LOCAL_RW_DATA 14 [11:0]											

PORT1 LOCAL-RW DATA 15 ~ PORT8 LOCAL RW DATA 15

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hE0~'hE7	0x0	x	x	x	x	PORT1_LOCAL_RW_DATA 15 [11:0] ~ PORT8_LOCAL_RW_DATA 15 [11:0]											

PORT1 LOCAL-RW DATA 16 ~ PORT8 LOCAL RW DATA 16

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hE8~'hEF	0x0	x	x	x	x	PORT1_LOCAL_RW_DATA 16 [11:0] ~ PORT8_LOCAL_RW_DATA 16[11:0]											

Bits	Fields	Descriptions
11:0	PORT _x _LOCAL_RW_DATA _y	These registers are used to contain the data to be written/read to/from the XD12 drivers daisied to each channel for accessing the XD12 drivers' registers. These registers should be written before executing the relevant commands, the local-reading or local-writing command. It needs to be careful to use that these registers are differently mapped according to the CH_SEG [1:0] of the LOCAL WRITE/READ COMMAND registers CH_SEG = 2'b00 : PORT1 to PORT8 are mapped to the channel 1 to 8 in order. CH_SEG = 2'b01 : PORT1 to PORT8 are mapped to the channel 9 to 16 in order. CH_SEG = 2'b10 : PORT1 to PORT8 are mapped to the channel 17 to 24 in order. The data order to be transferred to the XD12 drivers daisied in a channel_x : PORT_x_LOCAL_RW_DATA₁ to the 1st driver, PORT_x_LOCAL_RW_DATA₂ to the 2nd driver, The data order to be received from the XD12 drivers daisied in a channel_x : PORT_x_LOCAL_RW_DATA₁ from the last driver, PORT_x_LOCAL_RW_DATA₂ from the (last -1) driver, Refer to the Fig 10. for detail.

Soldering Temperature Reflow Profile

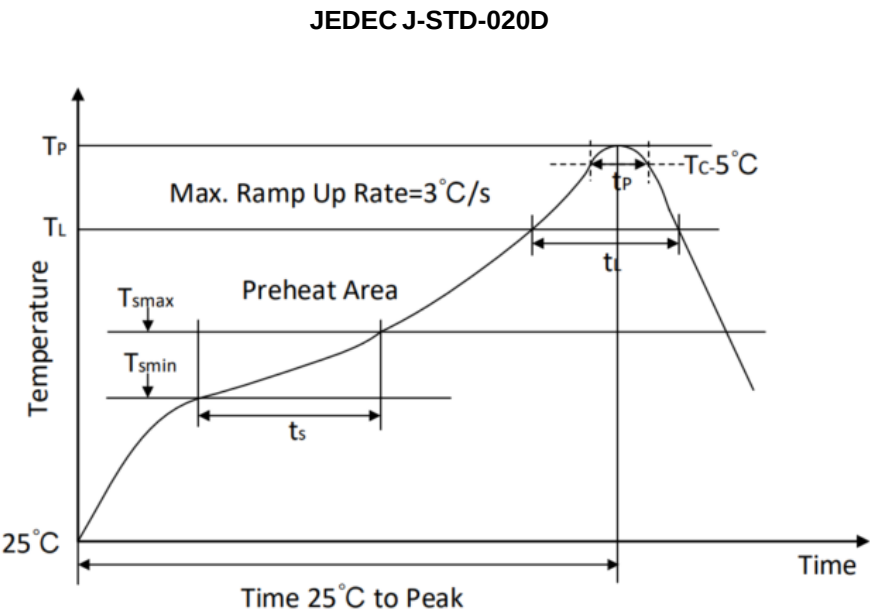


Fig 21. Reflow Soldering information

Table 9. Reflow Soldering Parameter

Parameter	Pb-free soldering conditions
Average ramp-up rate (217 °C to Peak)	3°C/second max.
Time of Preheat temp (from 150 °C to 200 °C)	60 ~ 120 second
Time to be maintained above 217 °C	60 ~ 150 second
Peak temperature	260 +5/-0 °C
Time within 5 °C of actual peak temp	30 second
Ramp-down rate	6 °C/second max.
Time from 25°C to peak temp	8 minutes max.

Important statement

Global Technologies reserves the right to change the above-mentioned information without prior notice.

Package specification

QFN 6060-48L Package Outline

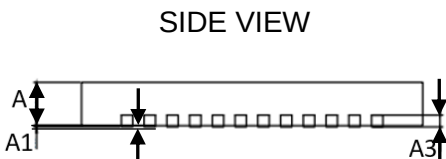
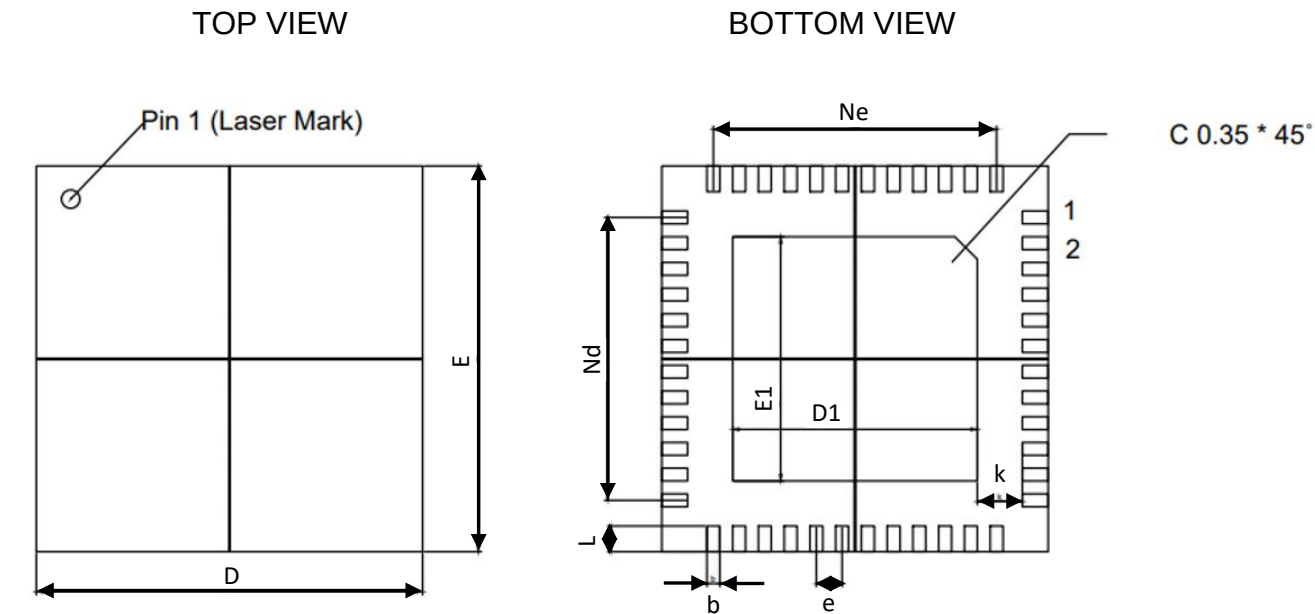
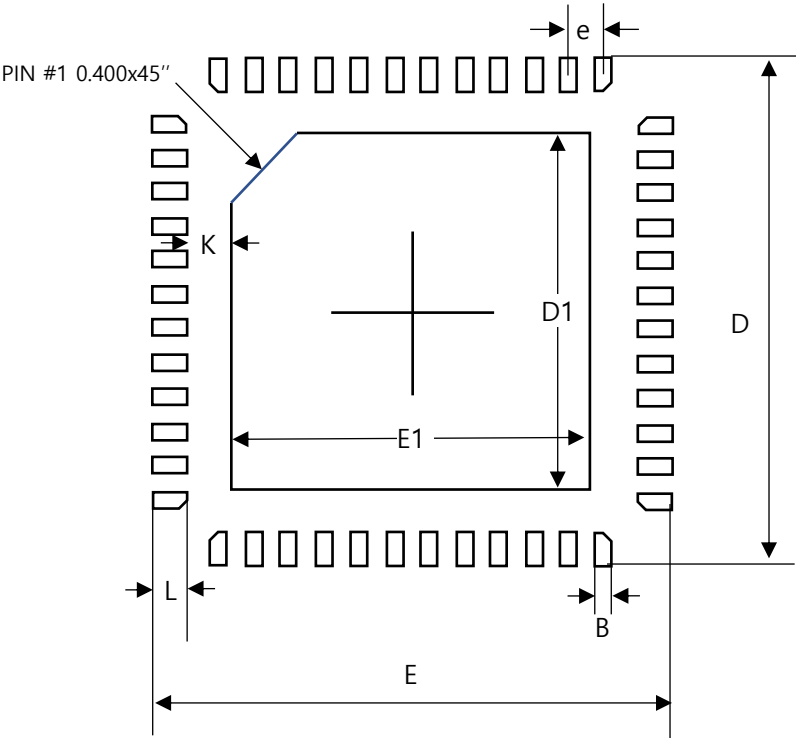


Fig 22. Package Outline

Symbol	Dimensions in mm		
	Min	Typ	Max
A	0.700	0.750	0.800
A1	0.000	0.020	0.050
A3	0.203 REF		
D	5.900	6.000	6.100
E	5.900	6.000	6.100
D1	4.3	4.4	4.5
E1	4.3	4.4	4.5
k	0.250 REF		
b	0.140	0.200	0.250
e	0.400 BSC		
Nd	4.400 BSC		
Ne	4.400 BSC		
L	0.350	0.400	0.450

Footprint



Symbol	Dimensions in mm		
	Min	Nom	Max
A	0.7	0.75	0.8
A1	0.00	0.02	0.05
A2	0.203 REF		
B	0.25	0.26	0.27
D		6.4	
D1	4.6	4.7	4.8
E1	4.6	4.7	4.8
E		6.4	
e	0.4 BSC		
K	0.25 REF		
L		0.60	

Fig 23. Recommended Footprint

Note: The reflow peak soldering temperature is peak 260°C and it is specified based on IPC/JEDEC J-STD- 020 “Moisture/Reflow Sensitivity Classification for Non-Hermetic Solid State Surface Mount Devices”

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